

Algebraic Geometry

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Based on the book “The Rising Sea: Foundations of Algebraic Geometry” by Ravi Vakil

Contents

1	Sheaves	1
1.1	Definitions	1
1.2	The category of sheaves	4
1.3	Sheafification	7
1.4	B-Sheaves	11
1.5	Abelian categories of sheaves	13
1.6	Adjunctions of sheaves	17
2	Schemes	21
2.1	Affine schemes	21
2.2	The structure sheaf on an affine scheme	27
2.3	Schemes	30
2.4	Projective schemes	37
2.5	Properties of schemes	42
2.5.1	Topological properties	42
2.5.2	Reducedness and integrality	45
2.5.3	Affine-locality	47
2.5.4	Normality and factoriality	50

1 Sheaves

1.1 Definitions

Definition 1.1.1

For a topological space X , its set of open subsets $\text{open}(X)$ is partially ordered by inclusion, and thus forms a small category. It is also a cofiltered category: for every $U, V \in \text{open}(X)$ they have a lower bound $U \cap V$.

Definition 1.1.2

Let X be a topological space, and $\mathcal{C}$ a category. A **$\mathcal{C}$ -valued presheaf on X** is a contravariant functor $\mathcal{F}: \text{open}(X)^{\text{op}} \rightarrow \mathcal{C}$. It is customary to denote $\mathcal{F}(U \subseteq V): \mathcal{F}(V) \rightarrow \mathcal{F}(U)$ by res_U^V . When $\mathcal{C}$ is a concrete category (i.e. its objects are sets), elements of $\mathcal{F}(U)$ are called **sections of $\mathcal{F}$ over U** , and for a section $s \in \mathcal{F}(V)$ we write $s|_U$ for $\text{res}_U^V(s)$.

A **morphism of presheaves** is a natural transformation.

Denote the category of $\mathcal{C}$ -valued presheaves on X by $\text{Psh}_{\mathcal{C}}(X)$.

Example 1.1.3

Let E be a set (or a topological space, or any mathematical object which lies in the same category as X and its open subsets). We can then define the presheaf of functions on E by $\mathcal{F}(U) = \text{hom}(U, E)$: the set of all functions $U \rightarrow E$. The restriction maps are very literally restriction: for $U \subseteq V$ and $f: V \rightarrow E$, we restrict f to U to obtain a function $f|_U: U \rightarrow E$.

Definition 1.1.4

Let $\mathcal{F}$ be a presheaf over X valued in a category $\mathcal{C}$ with filtered colimits, and $p \in X$. Then $\mathcal{U}(p) = \{U \subseteq \text{open} \mid p \in U\}$, the set of neighborhoods of p , is a cofiltered subcategory of $\text{open}(X)$, and so $\mathcal{F}$ restricts to a functor from $\mathcal{U}(p)^{\text{op}}$. Define $\mathcal{F}$'s **stalk** at p to be the filtered colimit

$$\mathcal{F}_p = \text{colim}_{U \in \mathcal{U}(p)} \mathcal{F}(U) \in \mathcal{C}.$$

For concrete categories, this is the set of **germs** of $\mathcal{F}$ at p :

$$\mathcal{F}_p = \{(U, s) \mid U \in \mathcal{U}(p), s \in \mathcal{F}(U)\},$$

modulo the relation where we identify (s, U) and (t, V) if there is a subset $p \in W \subseteq U, V$ such that $s|_W = t|_W$.

Taking colimits is functorial, so the correspondence $p \mapsto \mathcal{F}_p$ is functorial. Explicitly $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of sheaves over X valued in a concrete category, and $p \in X$ we define $\varphi_p: \mathcal{F}_p \rightarrow \mathcal{G}_p$ by

$$\varphi_p(U, s) = (U, \varphi_U(s)).$$

Definition 1.1.5

A presheaf $\mathcal{F}$ on X (valued in a concrete category) is a **sheaf** if it satisfies the **sheaf axiom**: for every

open subset $U \subseteq X$ and open cover $\{U_i\}$ of U and sections $s_i \in \mathcal{F}(U_i)$, if $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all i, j then there exists a *unique* $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$ for all i .

We can split the sheaf axiom in two:

- (1) the **identity axiom**: if $\{U_i\}$ is an open cover of U , and $s_1, s_2 \in \mathcal{F}(U)$ are such that $s_1|_{U_i} = s_2|_{U_i}$ for all i , then $s_1 = s_2$.
- (2) the **gluability axiom**: the sheaf axiom without requiring uniqueness.

We can also state the sheaf axiom categorically. For every open cover $\{U_i\}$ of U , we can consider the following diagram:

$$\mathcal{F}(U) \xrightarrow{\text{res}} \prod_i \mathcal{F}(U_i) \xrightleftharpoons[\text{res}_j]{\text{res}_i} \prod_{i,j} \mathcal{F}(U_i \cap U_j),$$

where $\mathcal{F}(U) \xrightarrow{\text{res}} \prod_i \mathcal{F}(U_i)$ is the map $s \mapsto (s|_{U_i})_i$, res_1 and res_2 are the maps

$$\text{res}_1(s_i)_i = (s_i|_{U_i \cap U_j})_{i,j}, \quad \text{res}_2(s_i)_i = (s_j|_{U_i \cap U_j})_{i,j}.$$

Categorically, res_1 's coordinate to $\mathcal{F}(U_i \cap U_j)$ composes the i th projection with $\text{res}_{U_i \cap U_j}^{U_i}$, and res_2 's coordinate composes the j th projection with $\text{res}_{U_i \cap U_j}^{U_j}$. These maps are defined for any presheaf, and we can see that $\mathcal{F}$ is a presheaf iff this is an equalizer diagram.

Indeed, the equalizer of $\text{res}_1, \text{res}_2$ are all $(s_i)_i$ such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$, i.e. a family of sections which agree on the overlaps. And there is an $s \in \mathcal{F}(U)$ whose projections $s|_{U_i}$ are s_i when the gluing axiom is satisfied, and it is unique (so an embedding, and thus an equalizer) when the identity lemma is satisfied.

This allows us to generalize sheaves to any category $\mathcal{C}$ with arbitrary products.

Remark 1.1.6

Alternatively for an open cover $\mathcal{U} = \{U_i\}_{i \in I}$ of U , we can define a subcategory of $\text{open}(X)$ consisting of objects $U_i, U_i \cap U_j$ with morphisms $U_i \cap U_j \subseteq U_i$. Then $\mathcal{F}$ restricts to a functor $\mathcal{F}|_{\mathcal{U}}: \mathcal{U}^{\text{op}} \rightarrow \mathcal{C}$. Then the sheaf condition says that the cone $\mathcal{F}(U) \Rightarrow \mathcal{F}|_{\mathcal{U}}$ whose components are the restriction maps res , is a limiting cone.

The category of $\mathcal{C}$ -valued sheaves on X is denoted $\text{Sh}_{\mathcal{C}}(X)$, it is a full subcategory of $\text{Psh}_{\mathcal{C}}(X)$; meaning sheaf morphisms are just presheaf morphisms between sheaves. The stalk of a sheaf at a point is just the stalk of it as a presheaf.

Example 1.1.7

If $\mathcal{F}$ is a presheaf on X , and $U \subseteq X$ is open, we define $\mathcal{F}$'s **restriction** to U to be the presheaf $\mathcal{F}|_U$ on U defined for $V \subseteq U$ by $\mathcal{F}|_U(V) = \mathcal{F}(V)$ (it is just the restriction of the functor $\mathcal{F}$ to the subcategory $\text{open}(U)$). If $\mathcal{F}$ is a sheaf, its restriction is clearly a sheaf as well.

Example 1.1.8

Let X be a topological space, $p \in X$, and S a set (or a group, module, etc.). We define the **skyscraper sheaf** p_*S by

$$p_*S(U) = \begin{cases} S & \text{if } p \in U \\ * & \text{else} \end{cases},$$

where $*$ is a single-point set; the restrictions are given by the identities or the unique morphism $S \rightarrow *$. This is indeed a sheaf: if $p \in U$ and $\{U_i\}$ is a covering of U with $s_i \in \mathcal{F}(U_i)$ such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$

then for every $p \in U_i$ we have a single value $s \in S$ which is equal to all $s_i \in \mathcal{F}(U_i)$, this is the glued section. Otherwise $p \notin U$ and s_i are all the unique element in $*$, so s must be too.

Example 1.1.9

Let X be a topological space and S a set. The **constant presheaf** associated with S is the presheaf $\underline{S}_{\text{pre}}$ which maps all open sets U to S . This is a presheaf, but not a sheaf in general. For example if $X = \{p, q\}$ is the two point space with the discrete topology then take $s_p \neq s_q$; they cannot be glued to a global section on X .

If we instead define $\underline{S}(U)$ to be the set of locally constant functions $U \rightarrow S$, then this is a sheaf, called the **constant sheaf**. It is just the sheaf of continuous functions $\underline{S}(U) = C(U, S)$ with S given the discrete topology. The sheaf of continuous functions $C(\bullet, Y)$ is a sheaf in general (by the gluing lemma from topology).

Example 1.1.10

Let $\pi: X \rightarrow Y$ be a continuous map, and $\mathcal{F}$ a presheaf over X . Define the **pushforward** (or **direct image**) presheaf $\pi_*\mathcal{F}$ by

$$\pi_*\mathcal{F}(V) = \mathcal{F}(\pi^{-1}V),$$

for $V \subseteq Y$, and $\text{res}_W^V = \text{res}_{\pi^{-1}W}^{\pi^{-1}V}$. This is clearly a presheaf; if $\mathcal{F}$ is a sheaf it is also a sheaf. Indeed, if $V \subseteq Y$ is open and $\{V_i\}_i$ is a cover with $s_i \in \pi_*\mathcal{F}(V_i) = \mathcal{F}(\pi^{-1}V_i)$. Then the glue of these is unique, because $\mathcal{F}$ is a sheaf.

The skyscraper sheaf p_*S is the pushforward of the constant sheaf $\underline{S}$ by the inclusion $\{p\} \rightarrow X$.

An important property of the direct image sheaf is that it is functorial: if $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of presheaves over X , then $\pi_*\varphi: \pi_*\mathcal{F} \rightarrow \pi_*\mathcal{G}$ defined by $\pi_*\varphi_V = \varphi_{\pi^{-1}V}$ is a morphism of presheaves.

Note that if $\pi: X \rightarrow Y$ is continuous, $\mathcal{F}$ is a sheaf over X , and $\pi(p) = q$, then there is a natural morphism of stalks

$$(\pi_*\mathcal{F})_q \rightarrow \mathcal{F}_p.$$

Indeed, the left-hand side is

$$\text{colim}_{q \in V} \mathcal{F}(\pi^{-1}V) = \text{colim}_{p \in \pi^{-1}V} \mathcal{F}(\pi^{-1}V).$$

Now a cone under $\mathcal{F}$ defines a unique cone under $\pi_*\mathcal{F}$ simply by restricting, and since $\text{colim} \pi_*\mathcal{F}$ is the initial cone under $\pi_*\mathcal{F}$ there is a canonical map to this restricted cone, in particular for $\text{colim} \mathcal{F}$.

In more concrete terms, $(\pi_*\mathcal{F})_q$ has elements (s, V) for $s \in \mathcal{F}(\pi^{-1}V)$, which we then map to $(s, \pi^{-1}V) \in \mathcal{F}_p$.

Definition 1.1.11

If $\mathcal{O}_X$ is a sheaf of rings over a topological space X (i.e. a sheaf over X valued in the category of rings), then the pair $(X, \mathcal{O}_X)$ is called a **ringed space**; $\mathcal{O}_X$ is called the **structure sheaf**. Sections of $\mathcal{O}_X$ over $U \subseteq X$ are called **functions** on U , global sections are called **global functions**.

Definition 1.1.12

If $\mathcal{O}_X$ is a ringed space, an $\mathcal{O}_X$ -**module** is a sheaf of Abelian groups $\mathcal{F}$ over X such that for each $U \subseteq X$, $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$ -module, and all of these modules are well-behaved in that when $U \subseteq V$, the following diagram commutes:

$$\begin{array}{ccc}
\mathcal{O}_X(V) \times \mathcal{F}(V) & \xrightarrow{\text{mult}} & \mathcal{F}(V) \\
\text{res}_{V,U} \times \text{res}_{V,U} \downarrow & & \downarrow \text{res}_{V,U} \\
\mathcal{O}_X(U) \times \mathcal{F}(U) & \xrightarrow{\text{mult}} & \mathcal{F}(U)
\end{array}$$

The horizontal arrows are $\mathcal{O}_X(U)$'s action on $\mathcal{F}(U)$.

The commutative diagram means that

$$(s \cdot x)|_U = (s|_U) \cdot (x|_U),$$

for $s \in \mathcal{O}_X(V)$, $x \in \mathcal{F}(V)$.

Note that just as modules generalize the concept of an Abelian group (Abelian groups are $\mathbb{Z}$ -modules), $\mathcal{O}_X$ -modules generalize the concept of a sheaf of Abelian groups. Indeed, a sheaf of Abelian groups is just an $\mathbb{Z}$ -module. Every $\mathbb{Z}$ -module is a sheaf of Abelian groups by definition, and conversely a sheaf of Abelian groups has a unique $\mathbb{Z}$ -module structure where for $U \subseteq X$ connected we have $\mathbb{Z}(U) \cong \mathbb{Z}$ which has a unique multiplication on $\mathcal{F}(U) \in \mathbf{Ab}$. We extend this to arbitrary $U \subseteq X$ by gluing.

Proposition 1.1.13

If $\mathcal{F}$ is an $\mathcal{O}_X$ -module, then for each $p \in X$, $\mathcal{F}_p$ is an $\mathcal{O}_{X,p}$ -module.

Proof

The condition that $\mathcal{F}$ be an $\mathcal{O}_X$ -module means that multiplication is a natural map $\mathcal{O}_X \times \mathcal{F} \rightarrow \mathcal{F}$. Colimits are taken pointwise in functor categories (in particular for the product of functors), so this gives a natural map $\mathcal{O}_{X,p} \times \mathcal{F}_p \rightarrow \mathcal{F}_p$. The natural map is given by mapping $(s, U), (x, V)$ to $(sx, U \cap V)$, which is easily seen to give a module structure. ■

Example 1.1.14

Consider a smooth manifold X ; this defines a ringed space $(X, \mathcal{O}_X)$ where $\mathcal{O}_X$ is the sheaf of smooth real-valued functions: $\mathcal{O}_X(U) = C^\infty(U)$. These are all real algebras, so we could view this as an $\mathbb{R}$ -ringed space.

If $\pi: V \rightarrow X$ is a vector bundle over X , then consider the sheaf of smooth local sections of π :

$$\Gamma(U) = \{\sigma: U \rightarrow V \text{ smooth} \mid \pi \circ \sigma = \text{id}_U\}.$$

Then Γ is an $\mathcal{O}_X$ -module: it is a classical result that a local section defined on U can be multiplied by a smooth real-valued function defined on U .

1.2 The category of sheaves

As discussed, presheaves form a category for which sheaves form a full subcategory. We will denote the category of $\mathcal{C}$ -valued presheaves over a space X by $\mathcal{C}_X^{\text{pre}}$ (instead of $\mathbf{Psh}_{\mathcal{C}}(X)$), and we will denote the full subcategory of sheaves by $\mathcal{C}_X \subseteq \mathcal{C}_X^{\text{pre}}$.

Note that if $\mathcal{F}, \mathcal{G} \in \mathcal{C}_X$ are sheaves (or presheaves), then the hom-set of sheaf morphisms $\mathcal{F} \rightarrow \mathcal{G}$ is indeed a set, since X as a category is small (functor categories under small categories are locally small). Furthermore, if $\mathcal{C}$ is enriched over some monoidal category (in our case, usually $\mathbf{Ab}$) so is $\mathcal{C}_X$. This is because it is true in general for functor categories over enriched categories.

Definition 1.2.1

Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{C}$ -valued sheaves defined on X . Define their **sheaf-hom** to be the sheaf whose sections are sheaf morphisms,

$$\mathbf{Hom}(\mathcal{F}, \mathcal{G})(U) = \text{hom}_{\mathcal{C}_X}(\mathcal{F}|_U, \mathcal{G}|_U).$$

As discussed, if $\mathcal{C}$ is enriched over a monoidal category $\mathcal{V}$ then so is $\mathcal{C}_X$, and so $\mathbf{Hom}(\mathcal{F}, \mathcal{G})$ takes values in $\mathcal{V}$. In particular, if $\mathcal{F}, \mathcal{G}$ are sheaves of Abelian groups, then so is their sheaf-hom.

Proposition 1.2.2

The sheaf-hom is indeed a sheaf.

Proof

We first show it is a sheaf: if $U \subseteq V$ then a sheaf morphism $\mathcal{F}|_V \rightarrow \mathcal{G}|_V$ restricts to a sheaf morphism $\mathcal{F}|_U \rightarrow \mathcal{G}|_U$ in the obvious way: its components are the same but only taken on subsets of U . This is clearly a presheaf.

To show that it is a sheaf, let $\{U_i\}$ be an open cover for $U \subseteq X$, and $\alpha_i: \mathcal{F}|_{U_i} \rightarrow \mathcal{G}|_{U_i}$ be sheaf morphisms such that $\alpha_i|_{U_i \cap U_j} = \alpha_j|_{U_i \cap U_j}$ for all i, j . In order to glue them together for a sheaf morphism $\alpha: \mathcal{F}_U \rightarrow \mathcal{G}_U$ we need to define α_W for all $W \subseteq U$. For $s \in \mathcal{F}(W)$ consider $s_i = s|_{U_i} \in U_i \cap W$, and its image under α_i to $\mathcal{G}(U_i \cap W)$, call it t_i . Then since $U_i \cap W$ cover W and since α is a morphism as well as $\mathcal{G}$ being a sheaf, we can uniquely glue t_i together to get a $t \in \mathcal{G}(W)$; this will be the image of s .

It is not hard to see that this is the unique gluing of α_i , and is itself a sheaf morphism. $\blacksquare$

Note:

In the above proof we needed only for $\mathcal{G}$ to be a sheaf, $\mathcal{F}$ can be a presheaf.

Furthermore $\mathbf{Hom}$ forms a bifunctor

$$\mathbf{Hom}: \mathcal{C}_X^{\text{op}} \times \mathcal{C}_X \rightarrow \mathbf{Set}_X,$$

or V_X instead of $\mathbf{Set}_X$ when $\mathcal{C}$ is enriched over V .

Note that if $\mathcal{O}_X$ is a ringed space, it is also a module over itself. Thus if $\mathcal{F}$ is an $\mathcal{O}_X$ module, we can form its **dual**:

$$\mathcal{F}^\vee = \mathbf{Hom}_{\text{Mod}_{\mathcal{O}_X}}(\mathcal{F}, \mathcal{O}_X).$$

We now focus on sheaves of Abelian groups. It is a classical result that functor categories over Abelian categories are Abelian, thus $\mathbf{Ab}_X^{\text{pre}}$ is an Abelian category. That is to say, the hom-sets (not hom-sheaves) have an Abelian group structure: for $\varphi, \psi: \mathcal{F} \rightarrow \mathcal{G}$ we can define $\varphi + \psi$ and $-\varphi$ by $(\varphi + \psi)(U) = \varphi(U) + \psi(U)$ and $(-\varphi)(U) = -\varphi(U)$. And we can form kernels and cokernels so that all monomorphisms are kernels and all epimorphisms are cokernels.

As with all limits in functor categories, these are done coordinate-wise. That is to say, if we denote by $\ker_{\text{pre}}$ and $\text{coker}_{\text{pre}}$ the kernel and cokernel of Abelian presheaf morphisms, we have

$$(\ker_{\text{pre}} \varphi)(U) = \ker(\varphi(U)), \quad (\text{coker}_{\text{pre}} \varphi)(U) = \text{coker}(\varphi(U)).$$

Remark 1.2.3

Let I be an indexing category and $\mathcal{C}$ be a category with arbitrary pullbacks. It is easy to see that a $\mathcal{C}$ -morphism $f: X \rightarrow Y$ is monic iff the following square is a pullback:

$$\begin{array}{ccc}
X & \xrightarrow{\quad \text{id} \quad} & X \\
\text{id} \downarrow & & \downarrow f \\
X & \xrightarrow{\quad f \quad} & Y
\end{array}$$

Because the functor category $[I, \mathcal{C}]$ also has arbitrary pullbacks, a natural transformation $\alpha: \mathcal{F} \Rightarrow \mathcal{G}$ is monic iff a similar diagram is a pullback. Because limits in a functor category are taken coordinate-wise, this is iff $\alpha(i)$ is monic for all $i \in I$.

Thus a natural transformation is monic iff its coordinates are.

Similarly, if $\mathcal{C}$ has arbitrary pushouts, natural transformations in $[I, \mathcal{C}]$ are epic iff their coordinates are.

In lieu of this note, we see that $\mathbf{Ab}_X^{\text{pre}}$ forms an Abelian category. Indeed, if $\varphi: \mathcal{F} \Rightarrow \mathcal{G}$ is a monic morphism of presheaves, then $\varphi(U): \mathcal{F}(U) \Rightarrow \mathcal{G}(U)$ is a monic morphism of Abelian groups for all $U \subseteq X$. Thus $\varphi(U) = \ker \text{coker } \varphi(U)$ for all $U \subseteq X$ and thus $\varphi = \ker \text{coker } \varphi$ as desired. Similar for epic morphisms.

Many properties of $\mathbf{Ab}_X^{\text{pre}}$ are seen to be pointwise. For one, if we set an open set $U \subseteq X$, the functor $\mathbf{Ab}_X^{\text{pre}} \rightarrow \mathbf{Ab}$ mapping $\mathcal{F} \mapsto \mathcal{F}(U)$ is an exact functor. Indeed, it is additive and preserves kernels and cokernels because these are taken pointwise.

Furthermore, a sequence of presheaves

$$\mathcal{F}_1 \rightarrow \mathcal{F}_2 \rightarrow \cdots \rightarrow \mathcal{F}_n$$

is exact iff $\mathcal{F}_1(U) \rightarrow \cdots \rightarrow \mathcal{F}_n(U)$ is for all $U \subseteq X$. Again, this is an immediate consequence of kernels and cokernels (and thus images) being taken coordinate-wise.

Everything we have said holds for presheaves valued in any Abelian category $\mathcal{A}$ (recall that Abelian categories have all finite limits and colimits). But we are more interested in the category of sheaves, in which case this is less obvious.

Note that the constant presheaf $\mathbf{0}_{\text{pre}}$ is a sheaf, so we can denote it by $\mathbf{0}$ without confusion. In order to show that $\mathcal{A}_X$ is an Abelian category, we must first show that kernels and cokernels exist in it. The following proposition shows that the kernel of a sheaf morphism is the same as the kernel of a presheaf morphism, and thus we get the existence of kernels for free.

Proposition 1.2.4

If $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of sheaves, then $\ker_{\text{pre}} \varphi$ is a sheaf. Thus $\ker \varphi = \ker_{\text{pre}} \varphi$ forms a kernel in the category of sheaves.

Proof

We show this for $\mathcal{A} = \mathbf{Ab}$: if $\{U_i\}_i$ is an open cover of $U \subseteq X$ and $s_i \in \ker \varphi(U_i)$ are such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all i, j , then by gluing they form a section $s \in \mathcal{F}(U)$. And by naturality

$$\varphi(U)(s)|_{U_i} = \varphi(U_i)(s|_{U_i}) = \varphi(U_i)(s_i) = 0.$$

The unique gluing of $0 \in \mathcal{G}(U_i)$ is $0 \in \mathcal{G}(U)$, so that $\varphi(U)(s) = 0$, meaning $s \in \ker \varphi(U)$ as desired. $\blacksquare$

Remark 1.2.5

A generalization of the above proposition shows that limits taken in the category of presheaves of sheaves forms a sheaf. That is, if $\Phi: I \rightarrow \mathcal{C}_X$ is a functor, then consider the inclusion $\iota: \mathcal{C}_X \hookrightarrow \mathcal{C}_X^{\text{pre}}$. Then $\lim \iota \circ \Phi$ is a sheaf, so $\lim \Phi$ exists in $\mathcal{C}_X$.

Thus if $\mathcal{C}$ has arbitrary pullbacks, so too does $\mathcal{C}_X^{\text{pre}}$ and thus so too does $\mathcal{C}_X$. It then follows that a morphism

of sheaves is monic iff its monic as a morphism of presheaves (considering the pullback condition remark 1.2.3) iff each component is monic (again by remark 1.2.3).

Unfortunately we now run in to our first hurdle: presheaf cokernels are not sheaves.

Example 1.2.6

Consider $\mathbb{C}$ with its usual topology, and let $\mathcal{O}_{\mathbb{C}}$ be the sheaf of holomorphic functions. Let $\mathcal{F}$ be the presheaf of functions admitting a holomorphic logarithm. Consider the following exact sequence of presheaves

$$0 \rightarrow \underline{\mathbb{Z}} \hookrightarrow \mathcal{O}_{\mathbb{C}} \xrightarrow{\exp(2\pi i)} \mathcal{F} \rightarrow 0,$$

where the second map is $f \mapsto \exp(2\pi i f)$. This is indeed an exact sequence of presheaves: the first morphism is an embedding, and the second is epic (if $f = \exp(g)$ then $g/2\pi i \in \mathcal{O}_{\mathbb{C}}$ is an origin for f). The composition is zero, as $\exp(2\pi i n) = 0$, and if $\exp(2\pi i f) = 0$ then $f(x) \in \mathbb{Z}$ for all $x \in \mathbb{C}$. Because $\mathbb{Z}$ is discrete, this means $f \in \underline{\mathbb{Z}}$ as desired.

This shows that $\mathcal{F}$ is the cokernel of $\underline{\mathbb{Z}} \hookrightarrow \mathcal{O}_{\mathbb{C}}$: but it is not a sheaf! Indeed $\text{id}: \mathbb{C} \rightarrow 0 \rightarrow \mathbb{C}$ has a local logarithm at every point in its domain, but no global logarithm. Thus $\mathcal{F}$ fails the gluing axiom.

Thus for now we must wait to demonstrate that $\mathcal{A}_X$ is an Abelian category.

1.3 Sheafification

There is a natural map

$$\mathcal{F}(U) \rightarrow \mathcal{F}_p,$$

for every $p \in U$ given by mapping $s \in \mathcal{F}(U)$ to $(U, s) \in \mathcal{F}_p$. Together these assimilate into a map

$$\mathcal{F}(U) \rightarrow \prod_{p \in U} \mathcal{F}_p.$$

Proposition 1.3.1

The natural map above is injective when $\mathcal{F}$ is a sheaf.

Proof

Suppose that $s, t \in \mathcal{F}(U)$ such that $(U, s) = (U, t)$ in $\mathcal{F}_p$. Then there exists a neighborhood $p \in W_p \subseteq U$ such that $s|_{W_p} = t|_{W_p}$. If this is true for all p , then W_p form an open cover of U and by the identity axiom, $s = t$. ■

Definition 1.3.2

An element $(s_p)_p \in \prod_{p \in U} \mathcal{F}_p$ consists of **compatible germs** if for all $p \in U$, s_p has a representative $\tilde{s}_p \in \mathcal{F}(U_p)$ for $p \in U_p \subseteq U$ such that the germ of $\tilde{s}_p$ at $q \in U_p$ is s_q for all $q \in U_p$.

Equivalently, this says that there is an open cover $\{U_i\}$ of U and sections $f_i \in \mathcal{F}(U_i)$ such that if $p \in U_i$ then s_p is the germ of f_i at p . If s is a section of $\mathcal{F}$ over U , then $(s)_p$ consists of compatible germs.

Proposition 1.3.3

The natural map $\mathcal{F}(U) \rightarrow \prod_{p \in U} \mathcal{F}_p$'s image is precisely the compatible germs.

Proof

We already showed that every tuple in its image consists of compatible germs, now we show the converse. Suppose $(s_p)_p$ consists of compatible germs, so let $\tilde{s}_p \in \mathcal{F}(U_p)$ be a representation as in the definition. Then U_p cover U , and $\tilde{s}_p|_{U_p \cap U_q}$'s germ at q is s_q so that $\tilde{s}_p, \tilde{s}_q$ agree on a neighborhood of q and by symmetry in a neighborhood of p and q . ■

Proposition 1.3.4

Suppose $\mathcal{F}$ is a presheaf of sets and $\mathcal{G}$ a sheaf of sets, and $\varphi_1, \varphi_2: \mathcal{F} \rightarrow \mathcal{G}$ are two morphisms. Further suppose that φ_1, φ_2 induce the same map on each stalk; show that $\varphi_1 = \varphi_2$.

Proof

Consider the following diagram for $U \subseteq X$ open,

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\varphi_i} & \mathcal{G}(U) \\ \downarrow & & \downarrow \iota \\ \prod_{p \in U} \mathcal{F}_p & \xrightarrow{\prod \varphi_{i,p}} & \prod_{p \in U} \mathcal{G}_p \end{array}$$

for $i = 1, 2$. It is easily seen that this diagram commutes, and because $\varphi_{1,p} = \varphi_{2,p}$ for all $p \in U$ it follows that

$$\iota \circ \varphi_1 = \iota \circ \varphi_2$$

and since ι is injective by proposition 1.3.1, we get that $\varphi_1 = \varphi_2$ as desired. ■

Proposition 1.3.5

A morphism of sheaves of sets is injective iff it induces an injection on all stalks. Similarly it is an isomorphism iff it induces an isomorphism on all stalks.

Proof

If $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ is an isomorphism, then by functoriality $\varphi_p: \mathcal{F}_p \rightarrow \mathcal{G}_p$ is as well for all p . Similarly if it is an injection then $\varphi_p(s_p) = \varphi_p(t_p)$ then $\varphi(s)$ and $\varphi(t)$ are equal in a subset $V \subseteq U$ so that $\varphi(s|_V) = \varphi(t|_V)$ and by injectivity this means that $s|_V = t|_V$ and thus $s_p = t_p$.

Conversely consider the commutative diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\varphi} & \mathcal{G}(U) \\ \downarrow \iota_{\mathcal{F}} & & \downarrow \iota_{\mathcal{G}} \\ \prod_{p \in U} \mathcal{F}_p & \xrightarrow{\sim} & \prod_{p \in U} \mathcal{G}_p \end{array}$$

where the bottom horizontal arrow is the injection/isomorphism $\prod_{p \in U} \varphi_p$. Then $\iota_{\mathcal{G}} \circ \varphi$ is injective, and so φ is.

Now to show that φ is surjective, let $s \in \mathcal{G}(U)$. Then $\iota_{\mathcal{G}}(s) \in \prod_U \mathcal{G}_p$ consists of compatible germs, and mapping it back to $\prod_U \mathcal{F}_p$ we see that it still consists of compatible germs. Thus it lies in the image of $\iota_{\mathcal{F}}$ by proposition 1.3.3, and thus has a preimage in $\mathcal{F}(U)$. Because all of the maps are injective, this is the preimage of s in $\mathcal{F}(U)$ as desired. ■

Note:

It is not true that sheaves with isomorphic stalks are isomorphic: there may not be an isomorphism between the sheaves which induces these isomorphisms.

Theorem 1.3.6 (Sheafification)

The inclusion $\text{Set}_X \hookrightarrow \text{Set}_X^{\text{pre}}$ has a left adjoint, called **sheafification**: $\text{sh}: \text{Set}_X^{\text{pre}} \rightarrow \text{Set}_X$.

Proof

We need to show for any presheaf $\mathcal{F}$ there exists a sheaf $\mathcal{F}^{\text{sh}}$ and morphism $\text{sh}: \mathcal{F} \rightarrow \mathcal{F}^{\text{sh}}$ such that for every sheaf $\mathcal{G}$ and morphism $g: \mathcal{F} \rightarrow \mathcal{G}$ there exists a unique morphism $f: \mathcal{F}^{\text{sh}} \rightarrow \mathcal{G}$ such that the below diagram commutes:

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{\text{sh}} & \mathcal{F}^{\text{sh}} \\ & \searrow g & \downarrow \exists! f \\ & & \mathcal{G} \end{array}$$

We construct $\mathcal{F}^{\text{sh}}$ explicitly. For $U \subseteq X$ open, define

$$\mathcal{F}^{\text{sh}}(U) = \left\{ (f_p \in \mathcal{F}_p)_{p \in U} \mid \begin{array}{l} \text{for all } p \in U \text{ there exists a neighborhood } p \in V \subseteq U \text{ and } s \in \mathcal{F}(V) \\ \text{such that } s_q = f_q \text{ for all } q \in V \end{array} \right\},$$

where by s_q we mean the germ of s in $\mathcal{F}_q$. For $U \subseteq V$ there is an obvious map $\mathcal{F}^{\text{sh}}(V) \rightarrow \mathcal{F}^{\text{sh}}(U)$ mapping $(f_p)_{p \in V} \mapsto (f_p)_{p \in U}$; it is easily seen that $\mathcal{F}^{\text{sh}}$ is a presheaf.

Note that the construction of $\mathcal{F}^{\text{sh}}(U)$ is just the set of all families of compatible germs over U , which we discussed in the previous few results.

Now let $U \subseteq X$ be open and $\{U_i\}_i$ an open cover of U . Then if we have $(f_p^i)_{p \in U_i} \in \mathcal{F}^{\text{sh}}(U_i)$ for all i which are compatible, this means $f_p^i = f_p^j$ for all $p \in U_i \cap U_j$. Then we must define $(f_p)_{p \in U}$ by $f_p = f_p^i$ for any $i \in U_i$ for which p is a member of. This is well-defined precisely by compatibility, and it is a member of $\mathcal{F}^{\text{sh}}(U)$ because for $p \in U$ there is a $p \in U_i \subseteq U$ and then there is a $p \in V \subseteq U_i$ and $s \in \mathcal{F}(V)$ such that $s_q = f_q^i = f_q$ for all $q \in V$ as desired. Thus $\mathcal{F}^{\text{sh}}$ is a sheaf.

Now we must demonstrate that $\mathcal{F}^{\text{sh}}$ satisfies the universal property of a left adjoint. There is an obvious map $\text{sh}: \mathcal{F} \rightarrow \mathcal{F}^{\text{sh}}$ which maps $s \in \mathcal{F}(U)$ to $(s_p \in \mathcal{F}_p)_{p \in U} \in \mathcal{F}^{\text{sh}}(U)$ (this is a section because s satisfies the condition for it to be in $\mathcal{F}^{\text{sh}}(U)$). Let $g: \mathcal{F} \rightarrow \mathcal{G}$ be a morphism, and for $U \subseteq X$ consider the following diagram

$$\begin{array}{ccccc} \mathcal{F}(U) & \xrightarrow{\text{sh}} & \mathcal{F}^{\text{sh}}(U) & \xrightarrow{\iota_{\mathcal{F}}} & \prod_{p \in U} \mathcal{F}_p \\ & \searrow g & & & \downarrow \prod_p g_p \\ & & \mathcal{G}(U) & \xrightarrow{\iota_{\mathcal{G}}} & \prod_{p \in U} \mathcal{G}_p \end{array}$$

Note that the image of $\prod_p g_p \circ \iota_{\mathcal{F}}$ consists of compatible germs in $\prod_{p \in U} \mathcal{G}_p$ and so we can fill this diagram in with a unique arrow $f: \mathcal{F}^{\text{sh}}(U) \rightarrow \mathcal{G}(U)$. It is easy to see by uniqueness that these f assemble into a morphism $\mathcal{F}^{\text{sh}} \rightarrow \mathcal{G}$. ■

Note:

In the language of category theory, this theorem says that $\mathbf{Set}_X \subseteq \mathbf{Set}_X^{\text{pre}}$ is a **reflective subcategory**. Note that this proof works for other concrete categories like $\mathbf{Ab}$, $\mathbf{Grp}$, $\mathbf{Ring}$, etc. It can be shown that for sufficiently nice $\mathcal{C}$ (e.g. complete and cocomplete), sheafification exists for $\mathcal{C}_X \subseteq \mathcal{C}_X^{\text{pre}}$.

This justifies the fact that inclusion $\mathcal{C}_X \subseteq \mathcal{C}_X^{\text{pre}}$ preserves limits: it is a right adjoint. This explains why the presheaf kernel of a sheaf morphism is a sheaf kernel for example.

Proposition 1.3.7

The canonical morphism $\mathcal{F} \rightarrow \mathcal{F}^{\text{sh}}$ induces an isomorphism on stalks.

Proof

This is easily seen by an explicit description of $\mathcal{F}^{\text{sh}}$'s stalks. ■

Remark 1.3.8 (The space of sections of a presheaf)

Suppose $\mathcal{F}$ is a presheaf of sets on X . We will construct a topological space F along with a continuous map $\pi: F \rightarrow X$ as follows. As a set define F to be the disjoint unions of $\mathcal{F}$'s stalks:

$$F = \coprod_{p \in X} \mathcal{F}_p.$$

Now for $s \in \mathcal{F}(U)$ let $F_s = \{(p, s_p) \mid p \in U\} \subseteq F$; then the collection of all F_s where $s \in \mathcal{F}(U)$ for arbitrary $U \subseteq X$ forms a base for a topology on F . Indeed, for $s \in \mathcal{F}(U)$ and $t \in \mathcal{F}(V)$ we see that $F_s \cap F_t$ is the union of all F_r where $r = s|_W = t|_W$ for all $W \subseteq U \cap V$ where $s|_W = t|_W$.

The map $\pi: F \rightarrow X$ is the obvious projection $(p, s_p) \mapsto p$. For $U \subseteq X$ open we have that $\pi^{-1}U$ is the union of all F_s where $s \in \mathcal{F}(U)$ so that π is continuous. Furthermore π is an open map: for $s \in \mathcal{F}(U)$ it maps F_s to U . Moreover $\pi|_{F_s}$ is a bijection, so that π is a local homeomorphism.

The space F is called the **space of sections**, or **étalé space** of $\mathcal{F}$.

We can give an alternative construction for the sheafification of $\mathcal{F}$ as the sheaf of sections of $\pi: F \rightarrow X$ (that is for $U \subseteq X$ we define $\mathcal{F}^{\text{sh}}(U)$ to be the set of continuous maps $\sigma: U \rightarrow F$ where $\pi \circ \sigma = \text{id}_U$). Let us denote by $\mathcal{F}^+$ this new construction.

For $U \subseteq X$ open we define $\varphi: \mathcal{F}^{\text{sh}}(U) \rightarrow \mathcal{F}^+(U)$ by mapping $(f_p)_{p \in U}$ to $\sigma: U \rightarrow F$, $p \mapsto f_p$. Its inverse is given by mapping σ to $(\sigma(p))_p$.

Example 1.3.9

Let S be a set and consider the constant presheaf $\underline{S}_{\text{pre}}$. Then its sheafification is the constant *sheaf*: $(\underline{S}_{\text{pre}})^{\text{sh}} = \underline{S}$.

We can show this with the universal property: $\text{sh}: \underline{S}_{\text{pre}} \rightarrow \underline{S}$ is given by inclusion (constant maps are locally constant). Then for a morphism $g: \underline{S}_{\text{pre}} \rightarrow \mathcal{F}$ where $\mathcal{F}$ is a sheaf, we define $\bar{g}: \underline{S} \rightarrow \mathcal{F}$ by mapping a locally constant $f: U \rightarrow S$ to the gluing of $(\bar{g}(f|_{U_i}))_i$ (where $U_i \subseteq U$ are its connected components so that $f|_{U_i}$ is a constant).

Let $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves, then recall the following equivalences (see remark 1.2.5 and proposition 1.3.5):

- (1) φ is monic as a morphism of sheaves,
- (2) φ is monic as a morphism of presheaves,

- (3) $\varphi(U)$ is monic for all $U \subseteq X$,
- (4) φ_p is monic for all $p \in X$.

A similar equivalence does not hold true for epimorphisms. This boils down to the fact that inclusion $\mathcal{C}_X \subseteq \mathcal{C}_X^{\text{pre}}$ does not necessarily preserve colimits, and thus not epimorphisms. So while a presheaf morphism is epic iff its components are, the same does not hold for sheaf morphisms. But it is true that epimorphisms are determined by stalks.

Proposition 1.3.10

Let $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves. Then it is an epimorphism (in the category of sheaves) iff $\varphi_p: \mathcal{F}_p \rightarrow \mathcal{G}_p$ is an epimorphism for all $p \in X$.

Proof

If φ_p is epic for all p , then suppose we have two sheaf morphisms $\psi_1, \psi_2: \mathcal{G} \rightarrow \mathcal{H}$ such that $\psi_1 \circ \varphi = \psi_2 \circ \varphi$. Then taking stalks we see that for all p , $(\psi_1)_p \circ \varphi_p = (\psi_2)_p \circ \varphi_p$. Since φ_p is epic this means $(\psi_1)_p = (\psi_2)_p$ for all p and by proposition 1.3.4 this means $\psi_1 = \psi_2$ so that φ is epic.

Conversely let $p \in X$ and suppose there are maps $f, g: \mathcal{G}_p \rightarrow X$ such that $f \circ \varphi_p = g \circ \varphi_p$. We can define maps $\bar{f}, \bar{g}: \mathcal{G} \rightarrow p_*X$ by $\bar{f}_U(s) = f(s_p)$ for $p \in U$ and $\bar{f}_U = *$ otherwise. It is easy to see that $\bar{f} \circ \varphi = \bar{g} \circ \varphi$ and so $\bar{f} = \bar{g}$ which can easily be shown to mean $f = g$. ■

Note:

In remark 1.6.2 will show that there is an adjunction between taking stalks and the skyscraper sheaf: $\bullet_p \dashv p_* \bullet$. This was used in the above proof, and it is easy to see then how to construct this.

Example 1.3.11

We will show that an epimorphism of sheaves need not be epic on each of its components.

Let $\mathcal{O}_{\mathbb{C}}$ be the sheaf of holomorphic functions and $\mathcal{O}_{\mathbb{C}}^*$ be the sheaf of invertible (nowhere zero) holomorphic functions. These are sheaves of Abelian groups, and we then have maps of sheaves

$$0 \rightarrow \underline{\mathbb{Z}} \xrightarrow{\times 2\pi i} \mathcal{O}_{\mathbb{C}} \xrightarrow{\exp} \mathcal{O}_{\mathbb{C}}^* \rightarrow 1.$$

We will soon show that this is an exact sequence of sheaves of Abelian groups.

In particular $\exp: \mathcal{O}_{\mathbb{C}} \rightarrow \mathcal{O}_{\mathbb{C}}^*$ is an epimorphism as we can show directly. For any $z \in \mathbb{C}$ the morphism of stalks $\exp_z$ is surjective because every invertible holomorphic function locally has a logarithm. But $\exp$ is not an epimorphism on each of its components, for example $\text{id} \in \mathcal{O}_{\mathbb{C}}^*(\mathbb{C} - 0)$ has no global logarithm.

1.4 B-Sheaves

Oftentimes it is more natural to define a sheaf on X by defining its sections on a base of X , instead of on every open subset of X . Recall that a base of a topology is a collection of open subsets $\mathcal{B} \subseteq \text{open}(X)$ such that every open subset of X can be written as a union of sets from $\mathcal{B}$.

In the absence of an ambient topology, we may mean a base to mean a collection $\mathcal{B}$ of subsets of X such that the intersection of any two sets from $\mathcal{B}$ can be written as a union of elements from $\mathcal{B}$. If we are given an arbitrary collection S of subsets of X , they generate a minimal topology on X ; S is called a **subbase** of this topology. The topology generated by $\mathcal{B}$ consists of all finite intersections of elements from $\mathcal{B}$, and $\mathcal{B}$ is a base of this topology (in the first sense).

As $\mathcal{B}$ is a collection of subsets of X , we can view it as a poset under inclusion, and thus a subcategory of $\text{open}(X)$.

Definition 1.4.1 (B-Sheaves)

Let X be a topological space with a given base $\mathcal{B}$. Define a **$\mathcal{B}$ -presheaf** to be a functor $\mathcal{F}: \mathcal{B}^{\text{op}} \rightarrow \text{Set}$. This presheaf is a **$\mathcal{B}$ -sheaf** if it satisfies the following two axioms:

- (1) the **identity axiom**: if $B = \bigcup_i B_i$ is a basic covering and $f, g \in \mathcal{F}(B)$ such that $f|_{B_i} = g|_{B_i}$ for all i , then $f = g$.
- (2) the **gluing axiom**: if $B = \bigcup_i B_i$ is a basic covering and $f_i \in \mathcal{F}(B_i)$ are given for all i such that $f_i|_{B_k} = f_j|_{B_k}$ for all $B_k \subseteq B_i \cap B_j$, then there is an $f \in \mathcal{F}(B)$ such that $f|_{B_i} = f_i$ for all i .

We can rewrite the following axioms categorically as follows. For $B = \bigcup_i B_i$, the following diagram must be an equalizer map

$$\mathcal{F}(B) \xrightarrow{\varphi} \prod_i \mathcal{F}(B_i) \xrightarrow{\alpha} \prod_{i,j,k} \mathcal{F}(B_{i,j,k})$$

where $\{B_{i,j,k}\}_k$ ranges over all basic sets contained in $B_i \cap B_j$, and

$$\theta = (\text{res}_{B, B_i})_i, \quad \alpha = (\text{res}_{B_i, B_k} \circ \text{Proj}_i)_{i,j,k}, \quad \beta = (\text{res}_{B_j, B_k} \circ \text{Proj}_j)_{i,j,k}.$$

Thus we can define $\mathcal{B}$ -sheaves over any category $\mathcal{C}$ with enough limits.

Definition 1.4.2

Denote the category of $\mathcal{C}$ -valued $\mathcal{B}$ -presheaves by $\mathcal{C}_B^{\text{pre}}$, and the category of $\mathcal{C}$ -valued $\mathcal{B}$ -sheaves by $\mathcal{C}_B$.

Theorem 1.4.3

Let $\mathcal{B}$ be a base for a topology on X , and $\mathcal{F}$ a sheaf of sets on this base. Then there is a unique sheaf of sets $\mathcal{F}^{\text{ext}}$ on X extending $\mathcal{F}$.

This extends to an equivalence of categories $\text{ext}: \text{Set}_B \xrightarrow{\sim} \text{Set}_X$: res , where $\text{res}: \text{Set}_X \rightarrow \text{Set}_B$ is the obvious restriction functor.

Proof

For a $\mathcal{B}$ -sheaf we can define stalks much the same way as for a usual sheaf. For $p \in X$ define

$$\mathcal{F}_p = \text{colim}_{p \in B \in \mathcal{B}} \mathcal{F}(B).$$

Then define for $U \subseteq X$,

$$\mathcal{F}^{\text{ext}}(U) = \left\{ (f_p \in \mathcal{F}_p)_{p \in U} \mid \begin{array}{l} \text{for all } p \in U \text{ there is a } B \in \mathcal{B} \text{ with } p \in B \subseteq U, \\ \text{such that } s_q = f_q \text{ for all } q \in B \end{array} \right\}.$$

There are obvious restriction maps and it is easy to see that this is a sheaf. Furthermore it is easy to see that $\mathcal{F}^{\text{ext}}(B)$ is the image of $\mathcal{F}(B)$ under $\iota_{\mathcal{F}}$ which is an injection as for usual sheaves. $\blacksquare$

Remark 1.4.4

Suppose we have an open cover $X = \bigcup_i U_i$ and for each i a sheaf of sets $\mathcal{F}_i$ on U_i . Furthermore suppose we have isomorphisms

$$\varphi_{ij}: \mathcal{F}_i|_{U_i \cap U_j} \xrightarrow{\sim} \mathcal{F}_j|_{U_i \cap U_j}$$

such that φ_{ii} is the identity and

$$\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik}$$

for all i, j, k on $U_i \cap U_j \cap U_k$.

There then exists a unique sheaf $\mathcal{F}$ on X such that $\mathcal{F}|_{U_i}$ is isomorphic to $\mathcal{F}_i$.

This is because we can define a base $\mathcal{B}$ consisting of those open subsets contained in some U_i , i.e. $\mathcal{B} = \bigcup_i \text{open}(U_i)$. Then there is a unique way to extend the $\mathcal{F}_i$ s to a $\mathcal{B}$ -sheaf. For $U \subseteq U_i$ we define $\mathcal{F}(U) = \mathcal{F}_i(U)$ (this is well-defined because if $U \subseteq U_i \cap U_j$ then $\mathcal{F}_i(U) \cong \mathcal{F}_j(U)$), and the restriction maps are given by the restriction maps of $\mathcal{F}_i$ and the φ_{ij} . That is, for $U \subseteq U_i$ and $V \subseteq U_j$ where $U \subseteq V$ we define $\text{res}: \mathcal{F}_j(V) \rightarrow \mathcal{F}_i(U)$ by

$$\text{res} = \mathcal{F}_j(V) \xrightarrow{\text{res}_j} \mathcal{F}_j(U) \xrightarrow{\varphi_{ji}} \mathcal{F}_i(U).$$

The conditions on φ_{ij} ensure that this is a $\mathcal{B}$ -sheaf, and we can extend it to a sheaf on X via the above theorem.

1.5 Abelian categories of sheaves

Recall that for a sheaf of rings $\mathcal{O}_X$, an $\mathcal{O}_X$ -**module** is a sheaf of Abelian groups $\mathcal{F}$ such that each $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$ -module in a compatible way (definition 1.1.12). A sheaf of Abelian groups is a $\mathbb{Z}$ -module, so $\mathcal{O}_X$ -modules generalize the concept of sheaves of Abelian groups much like how modules generalize the concept of Abelian groups.

The goal of this section is to demonstrate that the category of $\mathcal{O}_X$ -modules (and in particular sheaves of Abelian groups) forms an Abelian category.

To do this we need to show that the category of $\mathcal{O}_X$ -modules has kernels and cokernels, and that monics are kernels and epics are cokernels. Firstly let us consider $\text{Mod}_{\mathcal{O}_X}^{\text{pre}}$, the category of presheaf $\mathcal{O}_X$ -modules. The definition here is the same as definition 1.1.12 except we require only that $\mathcal{F}$ be a presheaf.

Now $\text{Mod}_{\mathcal{O}_X}^{\text{pre}}$ is complete and cocomplete. If $\Phi: I \rightarrow \text{Mod}_{\mathcal{O}_X}^{\text{pre}}$ is a diagram, then there is an obvious forgetful functor $\text{Mod}_{\mathcal{O}_X}^{\text{pre}} \rightarrow \text{Ab}_X^{\text{pre}}$. Composing this with Φ we can take the co/limit, and then there is an obvious $\mathcal{O}_X$ -module structure on the resulting object (this is the same situation as with $\text{Mod}_R \rightarrow \text{Ab}$).

Then sheafification lifts to $\text{Mod}_{\mathcal{O}_X}^{\text{pre}} \rightarrow \text{Mod}_{\mathcal{O}_X}$. This is clear by considering $\mathcal{F}^{\text{sh}}(U)$ explicitly. Thus $\text{Mod}_{\mathcal{O}_X}$ is a reflective subcategory of $\text{Mod}_{\mathcal{O}_X}^{\text{pre}}$, and since the latter is co/complete so too is the former.

In particular the kernel of a $\text{Mod}_{\mathcal{O}_X}$ -morphism is the kernel of it viewed as a morphism in Ab_X^{pre} , and its cokernel is the sheafification of this morphism.

Recall that the stalk of a morphism in $\text{Mod}_{\mathcal{O}_X}$ is an $\mathcal{O}_{X,p}$ -module. Thus we view $\bullet_p$ as a functor

$$\bullet_p: \text{Mod}_{\mathcal{O}_X} \rightarrow \text{Mod}_{\mathcal{O}_{X,p}}.$$

When $\mathcal{O}_X = \mathbb{Z}$ then $\text{Mod}_{\mathcal{O}_X}$ is the same as Ab_X , and $\mathcal{O}_{X,p}$ is the same as $\mathbb{Z}$ so that $\text{Mod}_{\mathcal{O}_{X,p}} \cong \text{Ab}$. Thus all of the following results hold for sheaves of Abelian groups.

Lemma 1.5.1

Taking stalks preserves kernels and cokernels: for $p \in X$ and a morphism $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ of $\mathcal{O}_X$ -modules,

$$(\ker \varphi)_p \cong \ker(\varphi_p), \quad (\text{coker } \varphi)_p \cong \text{coker}(\varphi_p).$$

Proof

This is clear for cokernels, as the stalk at p amounts to taking a certain colimit, and colimits (such as cokernels) commute with colimits.

For kernels we can show this directly. We view both $(\ker \varphi)_p$ and $\ker(\varphi_p)$ canonically as subobjects of $\mathcal{F}_p$, in which case it is clear that there is an equality. $\blacksquare$

Lemma 1.5.2

In $\mathbf{Mod}_{\mathcal{O}_X}$, a morphism $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ is monic iff its stalks are monic, epic iff its stalks are epic, and an isomorphism iff its stalks are isomorphisms.

Note:

It doesn't matter if we view φ_p as existing in $\mathbf{Mod}_{\mathcal{O}_{X,p}}$ or $\mathbf{Set}$; a module morphism (in the ordinary sense) is a monomorphism/epimorphism/isomorphism iff it is injective/surjective/bijective.

Note that in lieu of this, a $\mathbf{Mod}_{\mathcal{O}_X}$ -morphism with any of the above properties has those properties in $\mathbf{Ab}_X$ and $\mathbf{Set}_X$.

Proof

First we note that stalks determine a morphism: if $\varphi_p = \psi_p$ for all $p \in X$ then $\varphi = \psi$. This is because the forgetful functor $U: \mathbf{Mod}_{\mathcal{O}_X} \rightarrow \mathbf{Set}_X$ is faithful, and commutes with taking stalks. So $U\varphi_p = U\psi_p$, which implies $U\varphi = U\psi$ by proposition 1.3.4, and thus $\varphi = \psi$.

Then the proofs of proposition 1.3.5 and proposition 1.3.10 are valid here. $\blacksquare$

This allows us to prove the desired result.

Theorem 1.5.3

$\mathbf{Mod}_{\mathcal{O}_X}$ is an Abelian category.

Proof

As already shown, $\mathbf{Mod}_{\mathcal{O}_X}$ is co/complete. Furthermore it is clearly a pre-additive category. We need only show that for a monic φ and an epic ψ ,

$$\varphi = \ker \operatorname{coker} \varphi, \quad \psi = \operatorname{coker} \ker \psi.$$

But this is true only if for all $p \in X$,

$$\varphi_p = \ker \operatorname{coker} \varphi_p, \quad \psi_p = \operatorname{coker} \ker \psi_p \tag{1}$$

(parentheses on the left-hand sides of the equations here are unnecessary, as per lemma 1.5.1). By lemma 1.5.2 φ_p, ψ_p are monic and epic respectively, so (1) holds as $\mathbf{Mod}_{\mathcal{O}_{X,p}}$ is Abelian. $\blacksquare$

Note:

It is actually the case that if $\mathcal{A}$ is an arbitrary (complete) Abelian category, then $\mathcal{A}_X$ is Abelian as well.

Lemma 1.5.4

Let φ be a $\mathbf{Mod}_{\mathcal{O}_X}$ -morphism, then $\operatorname{im} \varphi$ is the sheafification of the image presheaf of φ . In particular $(\operatorname{im} \varphi)_p = \operatorname{im}(\varphi_p)$.

Proof

We know that $\operatorname{im}_{\operatorname{pre}} \varphi = \ker_{\operatorname{pre}} \operatorname{coker}_{\operatorname{pre}} \varphi$, while $\operatorname{im} \varphi = \ker \operatorname{coker} \varphi$. Since $\operatorname{coker} = \operatorname{coker}_{\operatorname{pre}}^{\operatorname{sh}}$ and $\ker = \ker_{\operatorname{pre}}^{\operatorname{sh}} = \ker_{\operatorname{pre}} \operatorname{so}$

$$\operatorname{im} \varphi = \ker_{\operatorname{pre}}^{\operatorname{sh}} \operatorname{coker}_{\operatorname{pre}}^{\operatorname{sh}} \varphi = (\ker_{\operatorname{pre}} \operatorname{coker}_{\operatorname{pre}} \varphi)^{\operatorname{sh}} = (\operatorname{im}_{\operatorname{pre}} \varphi)^{\operatorname{sh}}$$

as desired.

The second point follows from stalks commuting with kernels and cokernels. ■

Corollary 1.5.5

The following sequence of $\operatorname{Mod}_{\mathcal{O}_X}$ -morphisms

$$\mathcal{F} \xrightarrow{\alpha} \mathcal{G} \xrightarrow{\beta} \mathcal{H}$$

is exact iff

$$\mathcal{F}_p \xrightarrow{\alpha_p} \mathcal{G}_p \xrightarrow{\beta_p} \mathcal{H}_p$$

is exact for all $p \in X$.

Proof

The former is exact iff $\operatorname{im} \alpha = \ker \beta$ iff $(\operatorname{im} \alpha)_p = (\ker \beta)_p$ for all $p \in X$, which happens iff $\operatorname{im} \alpha_p = \ker \beta_p$, i.e. if the latter sequence is exact. ■

An immediate consequence of this corollary is the following.

Corollary 1.5.6

Taking stalks is an exact functor $\bullet_p: \operatorname{Mod}_{\mathcal{O}_X} \rightarrow \operatorname{Mod}_{\mathcal{O}_{X,p}}$.

Example 1.5.7

For $U \subseteq X$, there is a functor $\Gamma_U: \mathcal{C}_X \rightarrow \mathcal{C}$ mapping a sheaf $\mathcal{F} \mapsto \mathcal{F}(U)$. It is easy to see that $\Gamma_U: \operatorname{Mod}_{\mathcal{O}_X} \rightarrow \operatorname{Mod}_{\mathcal{O}_X(U)}$ is a left-exact functor. That is, it preserves kernels. This is clear because the kernel of a morphism of sheaves $\mathcal{F} \rightarrow \mathcal{G}$ is just the presheaf kernel, which is taken coordinate-wise.

When $U = X$ we denote this by just $\Gamma = \Gamma_X$, and it is called the **global sections functor**. Note that there is an obvious restriction functor $\operatorname{res}: \mathcal{C}_X \rightarrow \mathcal{C}_U$ mapping a sheaf $\mathcal{F}$ to its restriction $\mathcal{F}|_U$. Then Γ_U is equal to the composition

$$\Gamma_U = \mathcal{C}_X \xrightarrow{\operatorname{res}} \mathcal{C}_U \xrightarrow{\Gamma} \mathcal{C}.$$

And it is easy to see that res is also exact.

It is not true that Γ_U is exact though. Consider the exact sequence

$$0 \rightarrow \underline{\mathbb{Z}} \xrightarrow{\times 2\pi i} \mathcal{O}_{\mathbb{C}} \xrightarrow{\exp} \mathcal{O}_{\mathbb{C}}^* \rightarrow 1$$

of example 1.3.11. As before, $\Gamma_U(\exp)$ is not surjective as not every nowhere zero holomorphic function has a global inverse. Take for example $U = \mathbb{C} - 0$ and the function id .

Note:

Note that the maps $\text{res}: \mathcal{C}_V \rightarrow \mathcal{C}_U$ for $U \subseteq V$ forms a presheaf $\text{open}(X)^{\text{op}} \rightarrow \mathbf{Cat}$. In fact it is a $\mathbf{Cat}$ -valued sheaf.

Note:

We will show soon that $\Gamma: \mathcal{C}_X \rightarrow \mathcal{C}$ is right-adjoint to the constant sheaf functor. And as we will see $\text{res}: \mathcal{C}_X \rightarrow \mathcal{C}_U$ is the inverse image sheaf of the inclusion $U \hookrightarrow X$, which in general is left-adjoint to the pushforward, but in the case of an open embedding it is a right-adjoint as well. Thus $\Gamma_U = \Gamma \circ \text{res}$ is a right-adjoint and thus preserves limits; in particular it is left-exact.

Example 1.5.8

As alluded to in the previous note, the pushforward functor $\pi_*: \mathcal{C}_X \rightarrow \mathcal{C}_Y$ associated to a continuous map $\pi: X \rightarrow Y$ is a right-adjoint, and thus left-exact. We can show left-exactness directly: for a morphism $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ in $\mathcal{C}_X$, the components of $\pi_*\varphi$ are $(\pi_*\varphi)_V = \varphi_{f^{-1}V}$ for $V \subseteq Y$. Because kernels are taken coordinate-wise:

$$(\ker \pi_*\varphi)_V = \ker(\pi_*\varphi)_V = \ker \varphi_{f^{-1}V} = (\ker \varphi)_{f^{-1}V} = (\pi_* \ker \varphi)_V$$

so that $\ker \pi_*\varphi = \pi_* \ker \varphi$ as desired.

Definition 1.5.9

Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{O}_X$ -modules. Then we form their **tensor product presheaf** to be the presheaf $\mathcal{F} \otimes_{\mathcal{O}_X}^{\text{pre}} \mathcal{G}$ given by

$$(\mathcal{F} \otimes_{\mathcal{O}_X}^{\text{pre}} \mathcal{G})(U) = \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}(U), \quad \text{res}_{V,U} = \text{res}_{V,U}^{\mathcal{F}} \otimes \text{res}_{V,U}^{\mathcal{G}}.$$

Their **tensor product** is the sheafification of the tensor product presheaf:

$$\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G} = (\mathcal{F} \otimes_{\mathcal{O}_X}^{\text{pre}} \mathcal{G})^{\text{sh}}.$$

Proposition 1.5.10

For a presheaf $\mathcal{F}$, $\bullet \otimes^{\text{pre}} \mathcal{F}$ is left adjoint to $\mathbf{Hom}(\mathcal{F}, \bullet)$ as functors $\mathbf{Mod}_{\mathcal{O}_X}^{\text{pre}} \rightarrow \mathbf{Mod}_{\mathcal{O}_X}^{\text{pre}}$.

If $\mathcal{F}$ is a sheaf then $\bullet \otimes \mathcal{F}$ is left adjoint to $\mathbf{Hom}(\mathcal{F}, \bullet)$ as functors $\mathbf{Mod}_{\mathcal{O}_X} \rightarrow \mathbf{Mod}_{\mathcal{O}_X}$.

Furthermore, $\mathbf{Hom}(\bullet, \mathcal{F})$ is a self-adjoint functor. That is, there is an adjunction $\mathbf{Hom}(\bullet, \mathcal{F})^{\text{op}}: \mathbf{Mod}_{\mathcal{O}_X}^{\text{op}} \rightleftarrows \mathbf{Mod}_{\mathcal{O}_X}: \mathbf{Hom}(\bullet, \mathcal{F})$.

Proof

If $\varphi: \mathcal{G} \otimes^{\text{pre}} \mathcal{F} \rightarrow \mathcal{H}$ is a morphism, then for all $U \subseteq X$, $\varphi_U: \mathcal{G}(U) \otimes \mathcal{F}(U) \rightarrow \mathcal{H}(U)$ corresponds to a map $\psi_U: \mathcal{G}(U) \rightarrow \mathbf{Hom}_{\mathcal{O}_X(U)}(\mathcal{F}(U), \mathcal{H}(U))$. This then defines a map $\widehat{\varphi}_U: \mathcal{G}(U) \rightarrow \mathbf{Hom}(\mathcal{F}, \mathcal{H})(U)$ which maps $x \in \mathcal{G}(U)$ to $\widehat{\varphi}_U(x): \mathcal{F}|_U \rightarrow \mathcal{H}|_U$ whose components on $V \subseteq U$ are ψ_V .

Conversely for a map $\psi: \mathcal{G} \rightarrow \mathbf{Hom}(\mathcal{F}, \mathcal{H})$, we get maps $\mathcal{G}(U) \rightarrow \mathbf{Hom}(\mathcal{F}|_U, \mathcal{H}|_U) \xrightarrow{\Gamma} \mathbf{Hom}(\mathcal{F}(U), \mathcal{H}(U))$ which then gives us a map $\widehat{\psi}_U: \mathcal{G}(U) \otimes \mathcal{F}(U) \rightarrow \mathcal{H}(U)$. This assembles into the desired map $\widehat{\psi}: \mathcal{G} \otimes^{\text{pre}} \mathcal{F} \rightarrow \mathcal{H}$.

Thus we have adjunctions

$$\mathbf{Mod}_{\mathcal{O}_X}^{\text{pre}} \xrightleftharpoons[\mathbf{Hom}(\mathcal{F}, \bullet)]{\otimes^{\text{pre}} \mathcal{F}} \mathbf{Mod}_{\mathcal{O}_X}^{\text{pre}} \xrightleftharpoons[\text{inc}]{\text{sh}} \mathbf{Mod}_{\mathcal{O}_X}$$

so that $\bullet \otimes \mathcal{F}$ is the left adjoint of $\mathbf{Hom}(\mathcal{F}, \bullet)$ as desired.

For the final point we note that the adjunctions above are natural in all variables, so that

$$\mathrm{hom}(\mathcal{H}, \mathbf{Hom}(\mathcal{G}, \mathcal{F})) \cong \mathrm{hom}(\mathcal{H} \otimes \mathcal{G}, \mathcal{F}) \cong \mathrm{hom}(\mathcal{H} \otimes \mathcal{G}, \mathcal{F}) \cong \mathrm{hom}(\mathcal{H}, \mathbf{Hom}(\mathcal{G}, \mathcal{F}))$$

as desired. ■

Corollary 1.5.11

For a sheaf $\mathcal{F}$, $\mathbf{Hom}(\mathcal{F}, \bullet)$ is left-exact and $\mathbf{Hom}(\bullet, \mathcal{F})$ is contravariant left-exact.

Proposition 1.5.12

If $\mathcal{F}, \mathcal{G}$ are $\mathcal{O}_X$ -modules then for $p \in X$,

$$(\mathcal{F} \otimes \mathcal{G})_p = \mathcal{F}_p \otimes \mathcal{G}_p.$$

Proof

Filtered colimits in module categories commute with tensor products so

$$(\mathcal{F} \otimes \mathcal{G})_p = \mathrm{colim}_{p \in U} (\mathcal{F}(U) \otimes \mathcal{G}(U)) = (\mathrm{colim}_U \mathcal{F}(U)) \otimes (\mathrm{colim}_U \mathcal{G}(U)) = \mathcal{F}_p \otimes \mathcal{G}_p. \quad \blacksquare$$

1.6 Adjunctions of sheaves

In the following section, $\mathcal{C}$ is a complete and cocomplete category (or at least has all filtered colimits) for which sheafification exists.

Recall that for a continuous map $\pi: X \rightarrow Y$ there is a functor $\pi_*: \mathcal{C}_X \rightarrow \mathcal{C}_Y$ mapping a sheaf $\mathcal{F}$ to the sheaf $(\pi_*\mathcal{F})(V) = \mathcal{F}(\pi^{-1}V)$ for $V \subseteq Y$. It turns out that this functor has a left adjoint.

Theorem 1.6.1

The pushforward functor $\pi_*: \mathcal{C}_X \rightarrow \mathcal{C}_Y$ has a left adjoint, called the **inverse image functor**, $\pi^{-1}: \mathcal{C}_Y \rightarrow \mathcal{C}_X$.

Proof

First we show that $\pi_*: \mathcal{C}_X^{\mathrm{pre}} \rightarrow \mathcal{C}_Y^{\mathrm{pre}}$ has a left adjoint. For $U \subseteq X$ and $\mathcal{G} \in \mathcal{C}_Y^{\mathrm{pre}}$ define

$$\pi_{\mathrm{pre}}^{-1}\mathcal{G}(U) = \mathrm{colim}_{V \supseteq \pi(U)} \mathcal{G}(V).$$

This is clearly functorial in U and thus defines a presheaf in $\mathcal{C}_X^{\mathrm{pre}}$. Furthermore it is also clearly functorial in $\mathcal{G}$, defining a functor $\mathcal{C}_Y^{\mathrm{pre}} \rightarrow \mathcal{C}_X^{\mathrm{pre}}$.

To show the adjunction, consider we have a morphism $\varphi: \pi^{-1}\mathcal{G} \rightarrow \mathcal{F}$, which has components

$$\varphi_U: \mathrm{colim}_{V \supseteq \pi(U)} \mathcal{G}(V) \rightarrow \mathcal{F}(U).$$

Then define $\widehat{\varphi}: \mathcal{G} \rightarrow \pi_*\mathcal{F}$ whose component on $V \subseteq Y$ is

$$\widehat{\varphi}_V = \mathcal{G}(V) \rightarrow \mathrm{colim}_{W \supseteq \pi(\pi^{-1}V)} \mathcal{G}(W) \xrightarrow{\varphi_{\pi^{-1}V}} \mathcal{F}(\pi^{-1}V),$$

where the first morphism comes from the colimit cone.

Conversely given $\psi: \mathcal{G} \rightarrow \pi_* \mathcal{F}$ then for $U \subseteq X$ and $V \supseteq \pi U$ there is a morphism

$$\mathcal{G}(V) \xrightarrow{\psi_V} \mathcal{F}(\pi^{-1}V) \xrightarrow{\text{res}} \mathcal{F}(U).$$

These assemble into a cone under $(V \supseteq \pi U) \mapsto \mathcal{G}(V)$, and thus defines a map $\tilde{\psi}_U: \text{colim}_{V \supseteq \pi U} \mathcal{G}(V) \rightarrow \mathcal{F}(U)$. These assemble into a morphism $\tilde{\psi}: \pi^{-1} \mathcal{G} \rightarrow \mathcal{F}$.

Now for sheaves, we have adjunctions

$$\mathcal{C}_Y^{\text{pre}} \xrightleftharpoons[\pi_*]{\pi_{\text{pre}}^{-1}} \mathcal{C}_X^{\text{pre}} \xrightleftharpoons[\text{inc}]{\text{sh}} \mathcal{C}_X$$

so we define $\pi^{-1} = \text{sh} \circ \pi_{\text{pre}}^{-1}$ and this gives us the desired left adjoint. ■

Remark 1.6.2

A point $p \in X$ may be equivalently viewed as a morphism from the one-point space $p: * \rightarrow X$ which picks out p . Sheaves on $*$ amount to objects of $\mathcal{C}$ (because the sheaf condition implies $\mathcal{F}(\emptyset)$ is the terminal object), so that the inverse image-pushforward amounts to an adjunction

$$\mathcal{C}_X \xrightleftharpoons[p_*]{p^{-1}} \mathcal{C}.$$

When viewing $A \in \mathcal{C}$ as a sheaf over $*$, we get that $p_* A$ is the sheaf on X where for $U \subseteq X$,

$$(p_* A)(U) = A(p^{-1}U) = \begin{cases} A & p \in U \\ * & \text{else} \end{cases},$$

as A is the sheaf $A(*) = A$ and $A(\emptyset) = *$. This is precisely the skyscraper sheaf (which justifies the notation $p_* A$).

On the other hand, $p^{-1} \mathcal{F} \in \mathcal{C}$ is the colimit

$$\text{colim}_{U \ni p} \mathcal{F}(U) = \mathcal{F}_p$$

so that p^{-1} is the stalk functor at p .

So in this specific case, we have an adjunction $\bullet_p \dashv p_* \bullet$.

Proposition 1.6.3

The construction $\pi \mapsto \pi_*$ is functorial, and $\pi \mapsto \pi^{-1}$ is contravariantly functorial. That is, these form functors

$$\begin{aligned} \text{Top} &\rightarrow \text{Cat}, & X &\mapsto \mathcal{C}_X, & \pi &\mapsto \pi_* \\ \text{Top}^{\text{op}} &\rightarrow \text{Cat}, & X &\mapsto \mathcal{C}_X, & \pi &\mapsto \pi^{-1}. \end{aligned}$$

Proof

Because π^{-1} is the left adjoint of π_* , it is sufficient to show that one construction is functorial to show that the second is. We will show that if we have a composition of continuous maps $X \xrightarrow{\pi} Y \xrightarrow{\tau} Z$, then $(\tau \circ \pi)_* = \tau_* \circ \pi_*$. Indeed,

$$(\tau_* \circ \pi_* \mathcal{F})(W) = \pi_* \mathcal{F}(\tau^{-1}W) = \mathcal{F}((\tau \circ \pi)^{-1}W) = (\tau \circ \pi)_* \mathcal{F}(W). \quad \blacksquare$$

Note:

It should be made more clear what it is meant for $\pi \mapsto \pi^{-1}$ to be functorial. Indeed, it is not strictly true that $(\tau \circ \pi)^{-1} = \pi^{-1} \circ \tau^{-1}$; this equality is not strict and is rather a natural isomorphism. So $\pi \mapsto \pi^{-1}$ is not an ordinary functor, rather a 2-functor.

Corollary 1.6.4

Let $\pi: X \rightarrow Y$ be continuous, and $\mathcal{G} \in \mathcal{C}_Y$. Then for $p \in X$ there is a natural isomorphism $(\pi^{-1}\mathcal{G})_p \cong \mathcal{G}_{\pi p}$.

Proof

Note that $(\pi^{-1}\mathcal{G})_p$ is the composition $p^{-1} \circ \pi^{-1}(\mathcal{G}) = (\pi \circ p)^{-1}\mathcal{G}$, with $p: * \rightarrow X$. This is equal to $\pi(p)^{-1}(\mathcal{G}) = \mathcal{G}_{\pi p}$ as desired. ■

Proposition 1.6.5

Suppose $U \subseteq X$ is an open subset and consider the inclusion $\iota: U \hookrightarrow X$. Then the functors ι^{-1} and $\bullet|_U$ (restriction) are isomorphic.

Proof

We will show that $\bullet|_U$ is a left adjoint to ι_* . A map $\varphi: \mathcal{F}|_U \rightarrow \mathcal{G}$ defines $\widehat{\varphi}: \mathcal{F} \rightarrow \iota_*\mathcal{G}$ by

$$\widehat{\varphi}_V = \mathcal{F}(V) \xrightarrow{\text{res}} \mathcal{F}(U \cap V) \xrightarrow{\varphi|_{U \cap V}} \mathcal{G}(U \cap V) = \iota_*\mathcal{G}(V).$$

Conversely $\psi: \mathcal{F} \rightarrow \iota_*\mathcal{G}$ defines $\widetilde{\psi}: \mathcal{F}|_U \rightarrow \mathcal{G}$ by $\widetilde{\psi}_V = \psi_V$ for $V \subseteq U$, as $\iota_*\mathcal{G}(V) = \mathcal{G}(V)$. ■

Proposition 1.6.6

Let $\pi: X \rightarrow Y$ be continuous, then $\pi^{-1}: \mathbf{Ab}_Y \rightarrow \mathbf{Ab}_X$ is exact.

Proof

Exactness can be checked on the level of stalks by corollary 1.5.5. Thus π^{-1} is exact iff $p^{-1} \circ \pi^{-1}$ is exact for all $p \in X$ (where p^{-1} denotes taking the stalk at p). But this is isomorphic to $\pi(p)^{-1} = \bullet_{\pi p}$ which is exact. ■

Consider the following situation: we have a commutative square of continuous maps

$$\begin{array}{ccc} W & \xrightarrow{\beta'} & X \\ \alpha' \downarrow & & \downarrow \alpha \\ Y & \xrightarrow{\beta} & Z \end{array}$$

For a sheaf $\mathcal{F}$ on X we can define the **push-pull map**

$$\beta^{-1} \circ \alpha_* \mathcal{F} \rightarrow \alpha'_*(\beta')^{-1} \mathcal{F}$$

as follows. Consider the unit of $(\beta')^{-1} \dashv \beta'_*$, $\eta^{\beta'}: \mathcal{F} \rightarrow (\beta'_*)(\beta')^{-1}\mathcal{F}$. Then apply α_* to get $\alpha_*\mathcal{F} \rightarrow \alpha_*(\beta'_*)(\beta')^{-1}\mathcal{F}$. By commutativity this is equal to $\alpha_*\mathcal{F} \rightarrow (\beta_*)(\alpha'_*)(\beta')^{-1}\mathcal{F}$, and we can apply the adjunction $\beta^{-1} \dashv \beta$ to get $\beta^{-1}\alpha_*\mathcal{F} \rightarrow \alpha'_*(\beta')^{-1}\mathcal{F}$ as desired.

Recall that given an adjunction $(\eta, \varepsilon): L \dashv R$, the isomorphism $\text{hom}(c, Rd) \xrightarrow{\sim} \text{hom}(Lc, d)$ is given $f \mapsto \varepsilon \circ Lf$. Thus the push-pull map is given by

$$\varepsilon_{\alpha'_*(\beta')^{-1}}^{\beta} \circ (\beta^{-1} \circ \alpha_*(\eta^{\beta'})).$$

But we could define this another way. Instead we start with the counit $\varepsilon^{\alpha}: \alpha^{-1}\alpha_*\mathcal{F} \rightarrow \mathcal{F}$, apply $(\beta')^{-1}$: $(\beta')^{-1}\alpha^{-1}\alpha_*\mathcal{F} \rightarrow (\beta')^{-1}\mathcal{F}$, which is equivalent to $(\alpha')^{-1}\beta^{-1}\alpha_*\mathcal{F} \rightarrow (\beta')^{-1}\mathcal{F}$, then we apply the adjunction $(\alpha')^{-1} \dashv \alpha'_*$ to get the desired $\beta^{-1}\alpha_*\mathcal{F} \rightarrow \alpha'_*(\beta')^{-1}\mathcal{F}$. In this case the map becomes

$$(\alpha'_* \circ (\beta')^{-1})(\varepsilon^{\alpha}) \circ \eta^{\alpha'}.$$

These two definitions are equivalent. Indeed, we can view both of these maps as the $\mathcal{F}$ -component of a certain natural transformation, given by the two paths in the following diagram (the whiskering in the diagram is left implicit):

$$\begin{array}{ccc} \beta^{-1}\alpha_*\beta'_*(\beta')^{-1} & \xrightarrow{\sim} & \beta^{-1}\beta_*\alpha'_*(\beta')^{-1} \\ \eta^{\beta'} \uparrow & & \downarrow \varepsilon^{\beta} \\ \beta^{-1}\alpha_* & & \alpha'_*(\beta')^{-1} \\ \eta^{\alpha'} \downarrow & & \uparrow \varepsilon^{\alpha} \\ \alpha'_*(\alpha')^{-1}\beta^{-1}\alpha_* & \xrightarrow{\sim} & \alpha'_*(\beta')^{-1}\alpha^{-1}\alpha_* \end{array}$$

The claim is that this diagram commutes, and it can be proven by filling out the diagram using the naturality of the units, counits, and the triangle identities.

The push-pull map is also called the **change of basis map**.

Definition 1.6.7

Let $\mathcal{F}$ be a sheaf of Abelian groups on X , and $s \in \mathcal{F}(X)$ a global section. Define its **support** to be

$$\text{supp } s = \{p \in X \mid s_p \neq 0 \text{ in } \mathcal{F}_p\}.$$

Proposition 1.6.8

The support of a global section $\text{supp } s$ is a closed subset of X .

Proof

Note that $p \notin \text{supp } s$ iff $s_p = 0$ in $\mathcal{F}_p$ iff there is a neighborhood $p \in U$ such that $s|_U = 0$ in $\mathcal{F}(U)$. But then for all $q \in U$ one has $s_q = 0$ for this reason, so $U \subseteq (\text{supp } s)^c$; so $\text{supp } s$ is closed. ■

Note:

The locus where a continuous function is nonzero is open, i.e. if f is a continuous function to an Abelian group then $\{p \in X \mid f(p) \neq 0\}$ is open. In contrast the previous proposition shows that the locus where the germ of a continuous function (i.e. the stalk of a sheaf of continuous functions) is nonzero is closed.

This is because the latter is the set of all points where s is nonzero *near* (i.e. in a neighborhood around) p , while the former is the set of all points where f is nonzero *at* p .

In the following sheaves are valued in a category with a terminal object (so not **Ring** for example).

Definition 1.6.9

The **support** of a sheaf $\mathcal{F} \in \mathcal{C}_X$ is the set $\text{supp } \mathcal{F}$ of $p \in X$ for which $\mathcal{F}_p$ is non-terminal in $\mathcal{C}$.

If $\mathcal{F}$ is a sheaf of groups, then

$$\text{supp } \mathcal{F} = \{p \in X \mid \mathcal{F}_p \neq 0\} = \bigcup_{s \in \Gamma(\mathcal{F})} \text{supp } s.$$

Proposition 1.6.10

Let $Z \subseteq X$ be a closed subset, and denote by $\iota: Z \hookrightarrow X$ the inclusion.

- (1) If $\mathcal{F}$ is a sheaf on Z then for $q \in X$,

$$(\iota_* \mathcal{F})_q = \begin{cases} \mathcal{F}_q & q \in Z \\ * & q \notin Z \end{cases}$$

where $*$ $\in \mathcal{C}$ is the terminal object.

- (2) If $\mathcal{F}$ is a sheaf on X with $\text{supp } \mathcal{F} \subseteq Z$ then the unit $\mathcal{F} \rightarrow \iota_* \iota^{-1} \mathcal{F}$ is an isomorphism.

Proof

For (1) note that

$$(\iota_* \mathcal{F})_q = \text{colim}_{q \in U} \mathcal{F}(U \cap Z).$$

If $q \in Z$ then this is the colimit over all neighborhoods of q in Z , so isomorphic to $\mathcal{F}_q$. Otherwise there is a neighborhood of q disjoint from Z as it is closed, and $\mathcal{F}(\emptyset) = *$.

For (2), let $p \in X$. If $p \in Z$ then by (1) and corollary 1.6.4

$$(\iota_* \iota^{-1} \mathcal{F})_p \cong (\iota^{-1} \mathcal{F})_p \cong \mathcal{F}_p.$$

This isomorphism is the stalk of the unit. And if $p \notin Z$ then $p \notin \text{supp } \mathcal{F}$ so that $\mathcal{F}_p$ is terminal and so is $(\iota_* \iota^{-1} \mathcal{F})_p$ by (1).

Thus the stalks of the unit are all isomorphisms, and thus the unit itself is an isomorphism. ■

2 Schemes

2.1 Affine schemes

In this section we briefly review the definition of an affine scheme.

Definition 2.1.1

Let A be a commutative ring; its **spectrum** is the topological space whose underlying set is the set of all

prime ideals of A . We denote it by $\text{Spec}(A)$, and for a prime ideal $\mathfrak{p} \leq A$ we denote the element of $\text{Spec}(A)$ by $[\mathfrak{p}] \in \text{Spec}(A)$.

For a subset $S \subseteq A$ we define its **vanishing set** to be

$$V(S) = \{[\mathfrak{p}] \in \text{Spec}(A) \mid S \subseteq \mathfrak{p}\}.$$

The collection of all such vanishing sets, $\{V(S)\}_{S \subseteq A}$, forms a topology of closed sets on $\text{Spec}(A)$. This topology is called the **Zariski topology**.

Notice that for a subset $S \subseteq A$, $V(S) = V((S))$ where $(S) \leq A$ is the ideal generated by S . Furthermore,

$$V(\emptyset) = V(0) = \text{Spec}(A), \quad V(A) = \emptyset, \quad \bigcap_{i \in I} V(S_i) = V\left(\bigcup_{i \in I} S_i\right),$$

and that for two ideals $I, J \leq A$ we have that $V(I) \cup V(J) = V(IJ)$. It follows that the Zariski topology is indeed a topology. Notice that V is order-reversing: if $S_1 \subseteq S_2$ then $V(S_2) \subseteq V(S_1)$.

Recall that if $\varphi: A \rightarrow B$ is a ring morphism then it pulls back prime ideals: if $\mathfrak{q} \leq B$ is prime, then so is $\varphi^{-1}\mathfrak{q} \leq A$. Thus Spec is functorial:

$$\text{Spec}: \text{Ring}^{\text{op}} \rightarrow \text{Top}, \quad \varphi \mapsto \varphi^*$$

where $\varphi^*: \text{Spec}(B) \rightarrow \text{Spec}(A)$ is $[\mathfrak{q}] \mapsto [\varphi^{-1}\mathfrak{q}]$. This is continuous: for $I \leq A$

$$(\varphi^*)^{-1}V_A(I) = \{[\mathfrak{q}] \in \text{Spec}(B) \mid I \subseteq \varphi^{-1}\mathfrak{q}\} = \{[\mathfrak{q}] \in \text{Spec}(B) \mid \varphi(I) \subseteq \mathfrak{q}\} = V_B(\varphi(I)).$$

If $I \leq A$ is an ideal then its **radical** is the ideal

$$\sqrt{I} = \{a \in A \mid a^n \in I \text{ for some } n \geq 0\} \leq A.$$

It is easy to see that $V(I) = V(\sqrt{I})$: if $I \leq \mathfrak{p}$ then if $f \in \sqrt{I}$ then $f^n \in I \leq \mathfrak{p}$ for some n and since $\mathfrak{p}$ is prime, $f \in \mathfrak{p}$. Furthermore it can be seen that $\sqrt{I}$ is the intersection of all prime ideals containing I :

$$\sqrt{I} = \bigcap_{I \leq \mathfrak{p}} \mathfrak{p} = \bigcap V(I).$$

Clearly the left-hand side is contained in the right. Conversely if $f \notin \sqrt{I}$ then $\sqrt{I}$ and $\{f^n\}_{n \geq 0}$ do not intersect, and so $(\sqrt{I})_f \leq A_f$ is a proper ideal. Thus it is contained in a prime ideal $(\sqrt{I})_f \leq \mathfrak{p}_f \leq A_f$ associated to a prime ideal $\mathfrak{p} \leq A$, so that $\sqrt{I} \leq \mathfrak{p}$. But $f \notin \mathfrak{p}$, so $f \notin \bigcap V(I)$.

We also define the **nilradical** of A be the radical of the zero ideal:

$$\text{nil}(A) = \sqrt{0} = \{a \in A \mid a^n = 0 \text{ for some } n \geq 0\}.$$

Thus we have that

$$\text{nil}(A) = \bigcap_{[\mathfrak{p}] \in \text{Spec}(A)} \mathfrak{p}.$$

We can also form a base for the Zariski topology of $\text{Spec}(A)$. For $S \subseteq A$ let

$$D(S) = V(S)^c = \{[\mathfrak{p}] \in \text{Spec}(A) \mid S \not\subseteq \mathfrak{p}\}$$

form the open sets of the Zariski topology. For $f \in A$ let

$$D(f) = D\{f\} = \{[\mathfrak{p}] \in \text{Spec}(A) \mid f \notin \mathfrak{p}\}.$$

It is clear that

$$D(S) = \bigcup_{f \in S} D(f)$$

so that the $D(f)$ form a base of open sets for the topology.

We can map ideals of A to closed subsets of $\text{Spec}(A)$, we can also map subsets of $\text{Spec}(A)$ to ideals of A . For $X \subseteq \text{Spec}(A)$ we define the ideal

$$I(X) = \bigcap_{[\mathfrak{p}] \in X} \mathfrak{p} \leq A.$$

This is also clearly order-reversing: if $X_1 \subseteq X_2$ then $I(X_2) \subseteq I(X_1)$.

Notice that for an ideal $I \leq A$,

$$I(V(I)) = \bigcap_{I \leq \mathfrak{p}} \mathfrak{p} = \sqrt{I}.$$

And conversely for a subset $X \subseteq \text{Spec}(A)$,

$$V(I(X)) = \overline{X},$$

the closure of X in the Zariski topology. Indeed: if $X \subseteq V(I)$ then for $[\mathfrak{p}] \in X$ we have $I \leq \mathfrak{p}$ so that $I \leq \bigcap_{[\mathfrak{p}] \in X} \mathfrak{p} = I(X)$. Thus $V(I(X)) \subseteq V(I)$, so that $V(I(X))$ is the smallest closed subset containing X ; i.e. $\overline{X}$. Thus V and I form order-reversing bijections

$$\{\text{radical ideals of } A\} \xrightleftharpoons[I(\bullet)]{V(\bullet)} \{\text{closed subsets of } \text{Spec}(A)\}.$$

It follows that $V(I) \subseteq V(J)$ iff $\sqrt{J} \leq \sqrt{I}$: applying $I(\bullet)$ gives us the left-to-right direction, and $V(\sqrt{I}) = V(I)$ gives the right-to-left direction. In particular $V(I) = V(J)$ iff $\sqrt{I} = \sqrt{J}$. This carries over to D as well obviously: $D(I) \subseteq D(J)$ iff $\sqrt{I} \leq \sqrt{J}$. In particular $D(f) \subseteq D(g)$ iff $\sqrt{f} \leq \sqrt{g}$ iff $f^n \in (g)$ for some $n \geq 0$, or equivalently g is invertible in A_f (g is invertible in A_f iff $(g)_f$ is unital, iff (g) intersects the localizing set, iff $f^n \in (g)$ for some $n \geq 0$).

Because $V(0) = \text{Spec}(A)$, $V(I) = \text{Spec}(A)$ iff $\sqrt{I} = \text{nil}(A)$ iff $I \subseteq \text{nil}(A)$. In particular $D(f) = \emptyset$ iff $f \in \text{nil}(A)$. On the other hand, $V(I) = \emptyset$ for an ideal I iff $I = A$, since every proper ideal is contained in a prime ideal. Thus $D(f) = \text{Spec}(A)$ iff f is a unit.

We now discuss some topological results regarding $\text{Spec}(A)$.

Definition 2.1.2

A topological space X is said to be

- (1) **quasicompact** if every open cover of X has a finite subcover. Explicitly, if $X = \bigcup_{i \in I} U_i$ for $U_i \subseteq X$, then there are $i_1, \dots, i_n \in I$ such that $X = \bigcup_{j=1}^n U_{i_j}$.
- (2) **connected** if X cannot be written as the disjoint union of two nonempty open sets. Equivalently, X has no clopen subsets (subsets which are simultaneously open and closed) other than X and $\emptyset$. Otherwise X is **disconnected**.
- (3) **irreducible** if X cannot be written as the union of two proper closed subsets. That is if $X = Y \cup Z$ for Y, Z closed then either $X = Y$ or $X = Z$. Equivalently the intersection of any two nonempty open subsets is nonempty.

Note:

The term “quasicompact” may be odd to you, as this is sometimes taken to be the definition of a “compact” space. For geometers, oftentimes a compact space is a space which is both quasicompact and Hausdorff.

Remark 2.1.3

A space X is quasicompact iff every collection of closed sets with the finite intersection property has nonempty intersection. That is, if $\{Z_i\}_{i \in I}$ is a collection of closed subsets of X such that $\bigcap_{i \in I_0} Z_i$ is nonempty for all finite $I_0 \subseteq I$, then $\bigcap_{i \in I} Z_i \neq \emptyset$.

Remark 2.1.4

It is immediate that a space X is irreducible iff all nonempty open subsets of X are dense.

It is true that $\text{Spec}(A)$ is always quasicompact. It is sufficient to consider basic closed sets. If $\{V(f_i)\}_{i \in I}$ has the finite intersection property, then

$$\bigcap_{i \in I} V(f_i) = V(\{f_i\}_{i \in I})$$

and if this is empty, then $(f_i \mid i \in I) = A$. So there are $i_1, \dots, i_n \in I$ such that $1 \in (f_{i_1}, \dots, f_{i_n})$ so that

$$\emptyset = V(f_{i_1}, \dots, f_{i_n}) = \bigcap_{j=1}^n V(f_{i_j}),$$

contradicting the finite intersection property.

Definition 2.1.5

An **idempotent** in a ring A is an element $f \in A$ such that $f^2 = f$. A **nontrivial idempotent** is an idempotent which is not 0 or 1. A ring A is **connected** if it has no nontrivial idempotents; otherwise it is **disconnected**.

Proposition 2.1.6

Let A be a ring. Then the following conditions are equivalent:

- (1) A is disconnected;
- (2) A is isomorphic to the product of two non-trivial rings;
- (3) $\text{Spec}(A)$ is disconnected.

In particular $\text{Spec}(A \times B) \cong \text{Spec}(A) \amalg \text{Spec}(B)$.

Proof

Suppose (1), then if $f \in A$ is a nontrivial idempotent let $g = 1 - f$; this is also a nontrivial idempotent. The ideal $A^f = (f)$ has a ring structure whose addition and multiplication is inherited from A , its identity is f . We define $\varphi: A \rightarrow A^f \times A^g$ by $a \mapsto (fa, ga)$. This is injective because $fa + ga = a$, and it is surjective because $fg = 0$ so $\varphi(fa + gb) = (fa, gb)$. Thus $A \cong A^f \times A^g$, and these are nontrivial because $f, g \neq 0$; so (1) implies (2).

Conversely if $A \cong B \times C$ then $(1, 0), (0, 1) \in A$ are nontrivial idempotents so A is disconnected. Thus (2) implies (1).

Now again suppose (2), so consider the ring $A \times B$. Then notice that $\text{Spec}(A \times B) = D(1, 0) \sqcup D(0, 1)$. Indeed:

$$D(1, 0) \cup D(0, 1) = D((1, 0), (0, 1)) = D(1) = \text{Spec}(A \times B), \quad D(1, 0) \cap D(0, 1) = D((1, 0) \cdot (0, 1)) = D(0) = \emptyset.$$

Thus (2) implies (3).

Notice that the projection maps $A \times B \rightarrow A, B$ induce $\text{Spec}(A) \amalg \text{Spec}(B) \rightarrow \text{Spec}(A \times B)$. It is easy to see that this maps $\text{Spec}(A)$ to $D(1, 0)$ and $\text{Spec}(B)$ to $D(0, 1)$, showing that this is an isomorphism.

Now assume (3), so $\text{Spec}(A) = V(I) \sqcup V(J)$ where I, J are nonzero (so $V(I), V(J)$ are proper subsets). This means that $V(I + J) = \emptyset$ so that $I + J = A$ and $V(IJ) = \text{Spec}(A)$ so that $IJ = 0$. Thus by the Chinese remainder theorem, $A \rightarrow A/I \times A/J$ is an isomorphism, proving (2). ■

Proposition 2.1.7

$\text{Spec}(A)$ is irreducible iff $\text{nil}(A)$ is prime.

Proof

If $\text{nil}(A)$ is not prime then there are $f, g \notin \text{nil}(A)$ such that $fg \in \text{nil}(A)$. So $V(f), V(g) \neq \text{Spec}(A)$ and $\text{Spec}(A) = V(fg) = V(f) \cup V(g)$; thus $\text{Spec}(A)$ is reducible as desired.

Conversely if $\text{Spec}(A)$ is reducible then suppose $\text{Spec}(A) = V(I) \cup V(J)$ for proper subsets, so I, J are not subsets of $\text{nil}(A)$. But then $V(IJ) = \text{Spec}(A)$ so $IJ \leq \text{nil}(A)$, so $\text{nil}(A)$ is not prime. ■

Corollary 2.1.8

If A is a domain then $\text{Spec}(A)$ is irreducible.

Note:

While it is true that $\text{Spec} \prod_{i=1}^n A_i \cong \prod_{i=1}^n \text{Spec} A_i$ for finite products, this does not hold for infinite products. There is a canonical map $\prod_{i \in I} \text{Spec} A_i \rightarrow \text{Spec} \prod_{i \in I} A_i$; and it is easy to see that this map is an embedding. But if all the A_i are nonzero and I is infinite, the left hand side is not compact (because $\{\text{Spec} A_i\}_{i \in I}$ does not have a finite subcover) and thus it cannot be an isomorphism.

Definition 2.1.9

Let X be a topological space. A **closed point** of X is a point $x \in X$ such that $\overline{\{x\}} = \{x\}$.

In a Hausdorff space, all points are closed. For $\text{Spec} A$, we know that $\overline{[\mathfrak{p}]} = V(\text{I}([\mathfrak{p}])) = V(\mathfrak{p})$. Thus $[\mathfrak{p}]$ is a closed point iff $\mathfrak{p}$ is not contained in any prime ideal other than itself; equivalently it is a maximal ideal. That is: $[\mathfrak{p}] \in \text{Spec} A$ is a closed point iff $\mathfrak{p} \leq A$ is maximal.

Definition 2.1.10

A subset of a topological space $Z \subseteq X$ is an **irreducible component** if it is irreducible and maximally so: it is not properly contained in an irreducible set.

Proposition 2.1.11

A subset $Z \subseteq X$ is irreducible iff $\overline{Z} \subseteq X$ is.

Proof

If Z is irreducible and $\overline{Z} = Y \cup W$ are closed, then $Z \subseteq Y \cup W$ so that $Z \subseteq Y$ or $Z \subseteq W$. Then $\overline{Z} = Y$ or W because they are closed in $\overline{Z}$ and thus in X . Conversely if $\overline{Z}$ is irreducible then if $Z = Y \cup W$, then $\overline{Z} = \overline{Y} \cup \overline{W}$ so $\overline{Z} = \overline{Y}$ or $\overline{W}$. Suppose the former, so $Z = \overline{Y} \cap Z = Y$ since $Y \subseteq Z$ is closed. ■

Proposition 2.1.12

Every point $p \in X$ is contained in a (not necessarily unique) irreducible component, and all irreducible components of X are closed.

Proof

The first claim is due to Zorn. Clearly $\{p\}$ is irreducible, so we can form the nonempty poset of irreducible sets containing p . For a chain of such irreducible sets $\{Z_i\}_{i \in I}$, we have that $Z = \bigcup_{i \in I} Z_i$ is irreducible. Indeed if $Z = Y \cup W$ are closed then suppose $Y \subsetneq Z$ then there is an $f \in Z_i$ such that $f \notin Y$. Because Z_i is irreducible we must have $Z_i \subseteq W$. Then for all $j \geq i$ we have that $Z_j \subseteq W$ for the same reason; then $Z = \bigcup_{j \geq i} Z_j = W$ so that Z is irreducible as desired.

The second claim follows from the previous proposition: if Z is an irreducible component then $\overline{Z}$ is irreducible as well and since $Z \subseteq \overline{Z}$ by maximality $Z = \overline{Z}$. ■

Definition 2.1.13

Let $x, y \in X$ be points in a topological space such that $x \in \overline{\{y\}}$. Then x is called a **specialization** of y , and y a **generalization** of x .

A point $p \in X$ is a **generic point** if $\overline{\{p\}} = X$; i.e. if it is a generalization of all the points of X .

In $\text{Spec } A$ because $\overline{[\mathfrak{p}]} = V(\mathfrak{p})$, $[\mathfrak{p}]$ is a generalization of $[\mathfrak{q}]$ iff $[\mathfrak{q}] \in V(\mathfrak{p})$ iff $\mathfrak{p} \leq \mathfrak{q}$. And $[\mathfrak{p}]$ is a generic point iff $V(\mathfrak{p}) = \text{Spec}(A)$ iff $\mathfrak{p} = \text{nil } A$, so that $\text{Spec } A$ has a generic point iff $\text{nil } A$ is prime iff $\text{Spec } A$ is irreducible.

Definition 2.1.14

A poset Σ has the **descending chain condition (DCC)** if any of the following two equivalent conditions are satisfied:

- (1) every subset $\Sigma' \subseteq \Sigma$ has a minimal element;
- (2) every chain of elements $p_1 \geq p_2 \geq p_3 \geq \dots$ eventually stabilizes: $p_n = p_{n+1} = p_{n+2} = \dots$ for some n onward.

Dually Σ has the **ascending chain condition (ACC)** if Σ^{op} has the DCC.

The equivalence of the conditions is clear: (1) implies (2) by defining $\Sigma' = \{p_n\}_n$ and taking the minimum. Conversely if $\Sigma' \subseteq \Sigma$ has no minimal element, then we can form an infinite decreasing chain.

Definition 2.1.15

A topological space X is **Noetherian** if its poset of closed subsets has the DCC (equivalently its poset of open subsets has the ACC).

There is a notion of **Noetherian induction**: if Σ has DCC and we have some property P (which one can view as a function $\Sigma \rightarrow \{0, 1\}$) such that $P(x)$ holds for all minimal elements of Σ , and for $y \in \Sigma$ non-minimal if $P(x)$ holds for all $x < y$ then $P(y)$. Then $P(x)$ holds for all $x \in \Sigma$. This is easy to prove: take Σ' to be the set of all $x \in \Sigma$ for which $P(x)$ does not hold.

Using Noetherian induction one can prove the following.

Proposition 2.1.16

If X is a Noetherian topological space, then every nonempty closed subset $Z \subseteq X$ can be uniquely expressed as a finite union $Z = Z_1 \cup \cdots \cup Z_n$ where the Z_i are all irreducible closed subsets, none containing any other.

Remark 2.1.17

It is clear that if A is a Noetherian ring (its poset of ideals has ACC) then $\text{Spec } A$ is a Noetherian space. The converse is not true: if $\text{Spec } A$ is a Noetherian space then the poset of A 's *radical* ideals has ACC.

2.2 The structure sheaf on an affine scheme

Definition 2.2.1

Let A be a ring, then we define a sheaf of rings $(\text{Spec } A, \mathcal{O}_{\text{Spec } A})$ as follows. For a basic open set $D(f)$, define $\mathcal{O}_{\text{Spec } A}(D(f))$ to be the localization of A at all functions which do not vanish outside of $V(f)$, i.e. all $g \in A$ such that $V(g) \subseteq V(f)$.

For $f \in A$ let $S = \{g \in A \mid V(g) \subseteq V(f)\} = \{g \in A \mid D(f) \subseteq D(g)\}$. Recall that $V(g) \subseteq V(f)$ iff $f^n \in (g)$ iff $g \in A_f$ is invertible. Thus S is the saturation of $\{f^n\}_{n \geq 0}$, so that $S^{-1}A = \mathcal{O}_{\text{Spec } A}(D(f))$ and A_f are naturally isomorphic.

If $D(f') \subseteq D(f)$ then there is an obvious map $\mathcal{O}_{\text{Spec } A}(D(f)) \rightarrow \mathcal{O}_{\text{Spec } A}(D(f'))$ as the right ring is a localization of the left. Taking these maps as the restriction maps, this makes $\mathcal{O}_{\text{Spec } A}$ into a presheaf of rings (or A -algebras). To show that $\mathcal{O}_{\text{Spec } A}$ is a sheaf, we will make use of the following lemmas.

Lemma 2.2.2

Let A be a ring and $f_1, \dots, f_n$ generate the unit ideal. Then the following sequence of A -module maps is exact

$$0 \rightarrow A \xrightarrow{\varphi} \prod_i A_{f_i} \xrightarrow{\psi} \prod_{i,j} A_{f_i f_j},$$

where $\varphi(a) = (a)_i$ and $\psi(a_i) = (a_i - a_j)_{i,j}$. (We write $a \in A_f$ for the localization $a = a/1$.)

Proof

First we show that φ is injective: if $a = 0$ in all A_{f_i} then there are $m_1, \dots, m_n$ such that $f_i^{m_i} a = 0$ for all $i = 1, \dots, n$. But because $f_i \in \sqrt{(f_j^{m_j})_j}$ for all i and $(f_1, \dots, f_n) = A$, we see that $(f_1^{m_1}, \dots, f_n^{m_n}) = A$ as well. Thus there are $g_1, \dots, g_n \in A$ such that $g_1 f_1^{m_1} + \cdots + g_n f_n^{m_n} = 1$; multiplying by a shows that $a = 0$. Next we show that $\psi \circ \varphi = 0$. Indeed:

$$\psi \circ \varphi(a) = \psi(a)_i = (a - a)_{i,j} = 0,$$

so that $\text{im } \varphi \subseteq \ker \psi$. Now suppose that $(a_i/f_i^{m_i})_i \in \ker \psi$, then by multiplying both the numerator and denominator by f_i^s , we can assume that all the m_i 's are equal to some m . Thus

$$0 = \psi(a_i/f_i^m)_i = (a_i/f_i^m - a_j/f_j^m)_{i,j}$$

so that there is an M where $(f_i f_j)^M (f_j^m a_i - f_i^m a_j) = 0$ for all i, j .

As before, the f_i^{M+m} generate the unit ideal, so there are $b_i \in A$ such that $\sum_i b_i f_i^{M+m} = 1$. Then define $a = \sum_i b_i f_i^M a_i$ so that

$$f_j^{M+m} a = \sum_i b_i (f_i f_j)^M f_j^m a_i = \sum_i b_i (f_i f_j)^M f_i^m a_j = f_j^M \sum_i b_i f_i^{M+m} a_j = f_j^M a_j.$$

Thus in A_f , $a = a_j/f_j^m$, so that $\varphi(a) = (a_i/f_i^m)_i$ as desired. ■

We will say that a presheaf on X **has the sheaf property at** U for an open $U \subseteq X$ if the sheaf axiom holds for all open covers of U . So a sheaf is a presheaf which has the sheaf property at every open subset of X . Furthermore we can say that a presheaf has the sheaf property at U **relative to** $\{U_i\}_i$ for a specified open cover $\{U_i\}_i$ of U if the sheaf condition holds specifically for $\{U_i\}_i$.

Lemma 2.2.3

If X is a quasicompact space, then a presheaf $\mathcal{F}$ has the sheaf property at X iff it has the sheaf property at X relative to all finite covers of X .

Proof

One direction is clear. Conversely let $\mathcal{F}$ be a presheaf with the sheaf property at X relative to all finite covers of X , and let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open cover of X . Then there is a finite subset $I_0 \subseteq I$ such that $\mathcal{U}_0 = \{U_i\}_{i \in I_0}$ covers X . To show that $\mathcal{F}(U) \Rightarrow \mathcal{F}|_{\mathcal{U}}$ is a limiting cone (see remark 1.1.6), we note that a cone over $\mathcal{F}|_{\mathcal{U}}$ induces a cone over $\mathcal{F}|_{\mathcal{U}_0}$. Because $\mathcal{F}(U) \Rightarrow \mathcal{F}|_{\mathcal{U}_0}$ is limiting, any morphism of cones $(C \Rightarrow \mathcal{F}|_{\mathcal{U}}) \rightarrow (\mathcal{F}(U) \Rightarrow \mathcal{F}|_{\mathcal{U}})$ must be unique.

To show existence, it is sufficient to show that a morphism of cones $(C \Rightarrow \mathcal{F}|_{\mathcal{U}_0}) \rightarrow (\mathcal{F}(U) \Rightarrow \mathcal{F}|_{\mathcal{U}_0})$ is also a morphism of cones over $\mathcal{F}|_{\mathcal{U}}$. If not, then there is a finite diagram which doesn't commute: there is a $V \in \mathcal{U}$ such that $C \rightarrow \mathcal{F}(V)$ is not equal to $C \rightarrow \mathcal{F}(U) \rightarrow \mathcal{F}(V)$. Adding V to $\mathcal{U}_0$ yields a finite cover of U , and since $\mathcal{F}$ has the sheaf property relative to *all* finite covers of X , this is a contradiction. ■

Theorem 2.2.4

$\mathcal{O}_{\text{Spec } A}$ is a sheaf on the distinguished base, thus determines a unique sheaf on the topological space $\text{Spec } A$.

Proof

We must show that if $D(f) = \bigcup_{i \in I} D(f_i)$ then the following sequence is exact

$$0 \rightarrow A_f \xrightarrow{\varphi} \prod_i A_{f_i} \xrightarrow{\psi} \prod_{i,j} A_{f_i f_j}$$

where φ, ψ are defined as above (recall that $D(f) \cap D(g) = D(fg)$). For a finite I , this follows from the lemma before the previous with A_f in place of A . For general I this follows from the above lemma, along with the fact that $D(f) \cong \text{Spec}(A_f)$ is quasicompact. ■

Note:

It is *not* the case that $\mathcal{O}_{\text{Spec } A}(U)$ is the localization of A at all $f \in A$ not vanishing outside of U .

For a ring A , we can embed Mod_A fully faithfully into $\text{Mod}_{\mathcal{O}_{\text{Spec } A}}$. For an A -module M , let $\widetilde{M} \in \text{Mod}_{\mathcal{O}_{\text{Spec } A}}$ be the presheaf where $\widetilde{M}(D(f))$ is the localization of M at all $g \in A$ which don't vanish outside of $V(f)$. For the same reason as before, $\widetilde{M}(D(f)) \cong M_f$. The restriction maps are similar to before, and the proof that $\widetilde{M}$ is a B -sheaf, and thus determines a sheaf, is very similar to above.

In particular, $\widetilde{A}$ is just $\text{Spec } A$.

Lemma 2.2.5

Let A be a ring and $S \subseteq A$ a multiplicative set. We can view S as a preorder whose objects are the $f \in S$, and there is an arrow $f \rightarrow g$ iff $D(f) \subseteq D(g)$. Then for an A -module M , $f \mapsto M_f$ forms a functor $S^{\text{op}} \rightarrow \text{Mod}_{S^{-1}A}$, and

$$S^{-1}M \cong \text{colim}_{f \in S^{\text{op}}} M_f.$$

Proof

It is sufficient to prove this for $M = A$, as S is a cofiltered category ($fg \rightarrow f, g$) so the colimit is filtered. Because tensor products commute with filtered colimits,

$$\text{colim } M_f \cong \text{colim}(A_f \otimes M) \cong (\text{colim } A_f) \otimes M \cong S^{-1}A \otimes M \cong S^{-1}M.$$

The following proof works just as well for M , but is clearer with A .

Let $\varphi: A \rightarrow B$ be a ring map such that $\varphi(S) \subseteq B^\times$. Then $\varphi(f) \in B^\times$ so that this induces unique maps $\psi_f: A_f \rightarrow B$ for all $f \in S$ such that the following diagrams commute:

$$\begin{array}{ccc} A & \xrightarrow{\varphi} & B \\ & \searrow \iota_f & \uparrow \psi_f \\ & & A_f \end{array}$$

By uniqueness this forms a cone $(f \mapsto A_f) \Rightarrow B$, and thus a unique map $\text{colim } A_f \rightarrow B$. So we have a commutative diagram

$$\begin{array}{ccc} A & \xrightarrow{\varphi} & B \\ & \searrow & \uparrow \\ & & \text{colim } A_f \end{array}$$

Because $1 \in S$ and $A = A_1$, this gives rise to the desired map. This also shows uniqueness of the map: a map $\text{colim } A_f \rightarrow B$ commuting with φ gives rise to maps $A_f \rightarrow B$ commuting with $A \rightarrow A_f$ (again because $A = A_1$), and by the universal property of A_f , they are unique. ■

Corollary 2.2.6

Let $M \in \text{Mod}_A$ be an A -module and $\widetilde{M} \in \text{Mod}_{\mathcal{O}_{\text{Spec } A}}$ be the induced $\mathcal{O}_{\text{Spec } A}$ -module. Then for $[\mathfrak{p}] \in \text{Spec } A$ there is a natural isomorphism between the stalk of $\widetilde{M}$ at $[\mathfrak{p}]$ and the localization $M_{\mathfrak{p}}$:

$$\widetilde{M}_{[\mathfrak{p}]} \cong M_{\mathfrak{p}}.$$

In particular

$$(\text{Spec } A)_{[\mathfrak{p}]} \cong A_{\mathfrak{p}}.$$

Proof

The lefthand side is

$$\widetilde{M}_{[\mathfrak{p}]} = \text{colim}_{[\mathfrak{p}] \in D(f)} \widetilde{M}(Df) = \text{colim}_{f \notin \mathfrak{p}} M_f,$$

which is $(A - \mathfrak{p})^{-1}M = M_{\mathfrak{p}}$ by the previous lemma. ■

Proposition 2.2.7

Let M be an A -module. Then the natural map

$$M \hookrightarrow \prod_{\mathfrak{p} \in \operatorname{Spec} A} M_{\mathfrak{p}}$$

is injective.

Proof

This map corresponds to the map

$$M = \widetilde{M}(\operatorname{Spec} A) \hookrightarrow \prod_{\mathfrak{p} \in \operatorname{Spec} A} \widetilde{M}_{\mathfrak{p}} \cong \prod_{\mathfrak{p} \in \operatorname{Spec} A} M_{\mathfrak{p}}$$

which is an injection by proposition 1.3.1. ■

Note:

Alternatively we could prove this directly: if $m \in M$ is equal to zero in $M_{\mathfrak{p}}$ for all $\mathfrak{p} \in \operatorname{Spec} A$. Define the ideal

$$I = \operatorname{ann}_A(m) = \{a \in A \mid am = 0\} \leq A.$$

If m were nonzero, then I is a proper ideal and thus contained in a prime ideal $\mathfrak{p}$. For $a \notin \mathfrak{p}$ then $a \notin I$ so $am \neq 0$, and thus $m \neq 0$ in $M_{\mathfrak{p}}$ a contradiction.

Note that $M \mapsto \widetilde{M}$ is functorial. Indeed, an A -module map $\varphi: M \rightarrow N$ defines maps

$$\varphi_f: \widetilde{M}(\operatorname{D} f) = M_f \rightarrow N_f = \widetilde{N}(\operatorname{D} f)$$

which assemble into a morphism $\widetilde{\varphi}: \widetilde{M} \rightarrow \widetilde{N}$.

Note that this functor is fully faithful. It is faithful because $\widetilde{\varphi}_{\operatorname{D} 1} = \varphi$. And it is full because if $\psi: \widetilde{M} \rightarrow \widetilde{N}$ is a morphism, then let $\varphi = \psi_{\operatorname{D} 1}$, and then $\psi = \widetilde{\varphi}$. Indeed, by naturality we must have commutative squares for all $f \in A$:

$$\begin{array}{ccc} M & \xrightarrow{\quad \varphi \quad} & N \\ \downarrow & & \downarrow \\ M_f & \xrightarrow{\quad \psi_{\operatorname{D} f} \quad} & N_f \end{array}$$

By the universal property of localization we must have $\psi_{\operatorname{D} f} = \varphi_f$ as desired. Thus $M \mapsto \widetilde{M}$ is indeed a fully faithful embedding $\mathbf{Mod}_A \rightarrow \mathbf{Mod}_{\operatorname{Spec} A}$.

Definition 2.2.8

Let M be an A -module and $m \in M$. Define its **support** to be

$$\operatorname{supp} m = \{[\mathfrak{p}] \in \operatorname{Spec} A \mid m_{\mathfrak{p}} \neq 0\},$$

which is the support of m viewed as a global section of $\widetilde{M}$. We similarly define

$$\operatorname{supp} M = \operatorname{supp} \widetilde{M} = \{[\mathfrak{p}] \in \operatorname{Spec} A \mid M_{\mathfrak{p}} \neq 0\}.$$

2.3 Schemes

Note that when $\pi: X \rightarrow Y$ is a homeomorphism, there is an isomorphism of categories $\pi: \text{open}(Y) \cong \text{open}(X)$ mapping $V \mapsto \pi^{-1}V$. This lifts to an isomorphism of categories $\pi_*: \mathcal{C}_X^{\text{pre}} \cong \mathcal{C}_Y^{\text{pre}}$ mapping a presheaf $\mathcal{F} \in \mathcal{C}_X^{\text{pre}}$ to precomposition with π : $\pi_*\mathcal{F}(V) = \mathcal{F}(\pi^{-1}V)$. This is the usual pushforward, and it then induces an isomorphism $\mathcal{C}_X \cong \mathcal{C}_Y$.

Because π^{-1} is its left adjoint, which is unique up to isomorphism, this means that π^{-1}, π_* form an adjoint equivalence. In particular, the adjunction preserves categorical properties of morphisms, e.g. a morphism $\mathcal{F} \rightarrow \pi_*\mathcal{G}$ is an isomorphism iff $\pi^{-1}\mathcal{F} \rightarrow \mathcal{G}$ is.

Definition 2.3.1

Let $\mathcal{C}$ be a category (with enough limits and colimits). A **$\mathcal{C}$ -space** is a pair $(X, \mathcal{O}_X)$ where X is a topological space and $\mathcal{O}_X$ is a $\mathcal{C}$ -valued sheaf on X . A morphism of $\mathcal{C}$ -spaces $(X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is a continuous map $\pi: X \rightarrow Y$ along with a map of sheaves $\varphi: \mathcal{O}_Y \rightarrow \pi_*\mathcal{O}_X$. Composition is given in the obvious manner: if $X \xrightarrow{\pi} Y \xrightarrow{\tau} Z$ are continuous, and $\varphi: \mathcal{O}_Y \rightarrow \pi_*\mathcal{O}_X$ and $\psi: \mathcal{O}_Z \rightarrow \tau_*\mathcal{O}_Y$ are morphisms, then define

$$(\tau, \psi) \circ (\pi, \varphi) = \mathcal{O}_Z \xrightarrow{\psi} \tau_*\mathcal{O}_Y \xrightarrow{\tau_*\varphi} (\tau \circ \pi)_*\mathcal{O}_X.$$

We denote the category of $\mathcal{C}$ -spaces by $\text{Space}_{\mathcal{C}}$.

Note that composition is indeed associative. If we have maps

$$(X, \mathcal{O}_X) \xrightarrow{(\pi, \varphi)} (Y, \mathcal{O}_Y) \xrightarrow{(\tau, \psi)} (Z, \mathcal{O}_Z) \xrightarrow{(\sigma, \chi)} (W, \mathcal{O}_W)$$

then we have compositions

$$\begin{aligned} ((\sigma, \chi) \circ (\tau, \psi)) \circ (\pi, \varphi) &= (\sigma \circ \tau, \sigma_*\psi \circ \chi) \circ (\pi, \varphi) = (\sigma \circ \tau \circ \pi, (\sigma \circ \tau)_*\varphi \circ \sigma_*\psi \circ \chi) \\ (\sigma, \chi) \circ ((\tau, \psi) \circ (\pi, \varphi)) &= (\sigma, \chi) \circ (\tau \circ \pi, \tau_*\varphi \circ \psi) = (\sigma \circ \tau \circ \pi, \sigma_*(\tau_*\varphi \circ \psi) \circ \chi) \end{aligned}$$

which are equal.

Note that by the adjunction $\pi^{-1} \dashv \pi_*$, to give a morphism $\mathcal{O}_Y \rightarrow \pi_*\mathcal{O}_X$ is to give a morphism $\pi^{-1}\mathcal{O}_Y \rightarrow \mathcal{O}_X$. In which case the composition of $\varphi: \pi^{-1}\mathcal{O}_Y \rightarrow \mathcal{O}_X$ and $\psi: \tau^{-1}\mathcal{O}_Z \rightarrow \mathcal{O}_Y$ is

$$(\tau, \psi) \circ (\pi, \varphi): (\tau \circ \pi)^{-1}\mathcal{O}_Z \xrightarrow{\pi^{-1}\psi} \pi^{-1}\mathcal{O}_Y \xrightarrow{\varphi} \mathcal{O}_X.$$

Definition 2.3.2 (Schemes)

A **ringed space** is a Ring-space; we write RSp for $\text{Space}_{\text{Ring}}$.

An **affine scheme** is a ringed space which is isomorphic to $(\text{Spec } A, \mathcal{O}_{\text{Spec } A})$ for some ring A . A **scheme** is a ringed space $(X, \mathcal{O}_X)$ which is locally affine: for every $p \in X$ there is an open neighborhood $p \in U$ such that $(U, \mathcal{O}_X|_U)$ is affine. The topology of a scheme is called the **Zariski topology**.

Schemes are ringed spaces, but we will see that not all morphisms of ringed spaces should be considered morphisms of schemes. Though we will see that an isomorphism of schemes as ringed spaces gives an isomorphism of rings (when we define what morphisms of schemes are).

We can define the **global sections functor**

$$\Gamma: \text{Space}_{\mathcal{C}}^{\text{op}} \rightarrow \mathcal{C}, \quad \Gamma(X, \mathcal{O}_X) = \mathcal{O}_X(X).$$

A morphism $(\pi, \varphi): (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ determines a map $\varphi: \mathcal{O}_Y \rightarrow \pi_*\mathcal{O}_X$, which has component $\varphi_Y: \mathcal{O}_Y(Y) \rightarrow \mathcal{O}_X(\pi^{-1}Y) = \mathcal{O}_X(X)$. Thus we define $\Gamma(\pi, \varphi) = \varphi_Y$, turning Γ into a functor.

If X is a topological space and $U \subseteq X$ an open set, we may write $\Gamma(U, \mathcal{O}_X)$ instead of $\Gamma(U, \mathcal{O}_X|_U)$. In general we may write $(U, \mathcal{O}_X)$ instead of $(U, \mathcal{O}_X|_U)$.

Note:

Note that the global sections functor on sheaves is covariant, while the global sections functor on $\mathcal{C}$ -spaces is contravariant.

Lemma 2.3.3

There is a natural isomorphism

$$(D(f), \mathcal{O}_{\text{Spec } A}) \cong (\text{Spec } A_f, \mathcal{O}_{\text{Spec } A_f}).$$

Proof

Consider the canonical map $\iota: A \rightarrow A_f$, which induces the continuous injective $\iota: \text{Spec}(A_f) \rightarrow \text{Spec}(A)$. Its image consists of all prime ideals not intersecting $\{f^k\}_{k \geq 0}$, equivalently not containing f , so $D(f)$. It is also an open map: $D_{A_f}(g/f^n)$ contains all prime ideals of A_f not containing g/f^n ; since f^n is a unit, this is just all prime ideals not containing g . Thus $\iota D_{A_f}(g/f^n) = D(g) \cap D(f) = D(fg)$ which is open. Thus ι forms a homeomorphism $\text{Spec}(A_f) \xrightarrow{\sim} D(f)$.

Now we need to show that there is an isomorphism $\mathcal{O}_{\text{Spec } A}|_{D(f)} \cong \iota_* \mathcal{O}_{\text{Spec } A_f}$. Note that for $D(g) \subseteq D(f)$ we have

$$\mathcal{O}_{\text{Spec } A}(D(g)) = A_g, \quad \iota_* \mathcal{O}_{\text{Spec } A_f}(D(g)) = \mathcal{O}_{\text{Spec } A_f}(\iota^{-1} D(g)) = \mathcal{O}_{\text{Spec } A_f}(D(g)) = (A_f)_g \cong A_g.$$

So there is indeed an isomorphism, as desired. $\blacksquare$

Corollary 2.3.4

If $(X, \mathcal{O}_X)$ is a scheme and $U \subseteq X$ is open, then $(U, \mathcal{O}_X|_U)$ is a scheme.

Proof

For an affine scheme this is due to the previous lemma: because the $D(f)$ form a base, for any $p \in U$ there is an $f \in A$ such that $p \in D(f) \subseteq U$. Then $(D(f), \mathcal{O}_X|_{D(f)}) \subseteq (U, \mathcal{O}_X|_U)$ is an open affine subscheme.

In general let $p \in U$, then take $p \in V \subseteq X$ an affine neighborhood. Then $p \in U \cap V \subseteq U$ is an open subset of an affine scheme and thus there is an affine neighborhood $p \in W \subseteq U \cap V$. $\blacksquare$

Definition 2.3.5

If $(X, \mathcal{O}_X)$ is a scheme and $U \subseteq X$ open, then $(U, \mathcal{O}_X|_U)$ is called an **open subscheme** of X . If U is also an affine scheme, then U is an **affine open subscheme** or an **affine open subset**.

It is clear that for a scheme X , its affine open subsets form a base for the topology.

Definition 2.3.6

Let $\mathcal{C}$ be a category, a subcategory $\mathcal{C}_0 \subseteq \mathcal{C}$ is said to be **replete** if it is closed under isomorphisms: if $c \in \mathcal{C}_0$ and $f: c \xrightarrow{\sim} d$ is an isomorphism then both d and f are in $\mathcal{C}_0$.

Let $\mathcal{C}_0 \subseteq \mathcal{C}$ be a replete category. A $(\mathcal{C}, \mathcal{C}_0)$ -space is a $\mathcal{C}$ -space $(X, \mathcal{O}_X)$ such that for every $p \in X$, the stalk $\mathcal{O}_{X,p}$ is in $\mathcal{C}_0$. A morphism of $(\mathcal{C}, \mathcal{C}_0)$ -spaces is a $\mathcal{C}$ -space morphism $(\pi, \varphi): (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ such that for

every $p \in X$ the morphism of stalks $\varphi_p: (\pi^{-1}\mathcal{O}_Y)_p \cong \mathcal{O}_{Y,\pi(p)} \rightarrow \mathcal{O}_{X,p}$ is in $\mathcal{C}_0$.

Note:

We can write $\varphi_p: \mathcal{O}_{Y,\pi(p)} \rightarrow \mathcal{O}_{X,p}$ more explicitly. For $\pi(p) \in V \subseteq Y$ open we have a map

$$\varphi_V: \mathcal{O}_Y(V) \rightarrow \mathcal{O}_X(\pi^{-1}V) \rightarrow \operatorname{colim}_{p \in U} \mathcal{O}_X(U) = \mathcal{O}_{X,p}.$$

These then assemble into a cone $\mathcal{O}_Y(\bullet) \Rightarrow \mathcal{O}_{X,p}$ which then limits to $\mathcal{O}_{Y,\pi(p)} \rightarrow \mathcal{O}_{X,p}$.

Definition 2.3.7

Define **LocRing** to be the category of local rings $(A, \mathfrak{m})$. A morphism $\varphi: (A, \mathfrak{m}) \rightarrow (B, \mathfrak{n})$ is a ring morphism $A \rightarrow B$ such that $\varphi(\mathfrak{m}) \subseteq \mathfrak{n}$.

A **locally ringed space** is a $(\text{Ring}, \text{LocRing})$ -space. We denote the category of locally ringed spaces by **LocRSp**.

Note:

Importantly, a locally ringed space is not a **LocRing**-space. In fact, generally a locally ringed space's sections will not be local rings.

A scheme which is isomorphic to the spectrum of a ring is called an **affine scheme**. We denote the full subcategory of affine schemes by **AffSch**.

Schemes are locally ringed spaces: if X is a scheme and $p \in X$ then p has an affine neighborhood. Since stalks can be taken locally, it is sufficient to show that $\operatorname{Spec} A$ is a locally ringed space for any ring A . And this is clear as the stalk of $\operatorname{Spec} A$ at $[\mathfrak{p}] \in \operatorname{Spec} A$ is isomorphic to $A_{\mathfrak{p}}$ which is local.

Definition 2.3.8

The **category of schemes** is the full subcategory of **LocRing** whose objects are schemes; it is denoted by **Sch**. The full subcategory of **Sch** consisting of affine schemes is denoted **AffSch**.

Note that if X, Y are locally ringed space which are isomorphic as ringed spaces, then they are isomorphic as locally ringed spaces. Indeed in general if X, Y are $(\mathcal{C}, \mathcal{C}_0)$ -spaces which are isomorphic as $\mathcal{C}$ -spaces, they are isomorphic as $(\mathcal{C}, \mathcal{C}_0)$ -spaces. This is because $\mathcal{C}_0$ is replete (contains all isomorphisms).

Note that Spec forms a contravariant functor $\text{Ring}^{\text{op}} \rightarrow \text{AffSch}$ which maps a ring A to its spectrum. Then for a ring morphism $\varphi: A \rightarrow B$ we have a continuous map $\varphi^*: \operatorname{Spec} B \rightarrow \operatorname{Spec} A$ mapping $\mathfrak{q} \leq B$ to $\varphi^{-1}\mathfrak{q}$. And $\varphi^\sharp: \operatorname{Spec} A \rightarrow \varphi_*^* \operatorname{Spec} B$ which for $f \in A$ maps $\varphi^\sharp_f: A_f \rightarrow \operatorname{Spec} B((\varphi^*)^{-1}D_B(f)) = B_{\varphi f}$. This is just the natural map induced from the universal $A \xrightarrow{\varphi} B \rightarrow B_{\varphi f}$.

This is clearly seen to be functorial.

Definition 2.3.9

If $(X, \mathcal{O}_X)$ is a locally ringed space, then for $p \in X$ denote the maximal ideal of the stalk $\mathcal{O}_{X,p}$ by $\mathfrak{m}_{X,p}$ or $\mathfrak{m}_p$. The **residue field** is $\kappa(p) = \mathcal{O}_{X,p}/\mathfrak{m}_p$. For a function f on an open subset $U \subseteq X$ has a value at every $p \in U$ denote by $f(p)$: it is the image of the germ of f at p under the quotient $\mathcal{O}_{X,p} \rightarrow \kappa(p)$. We say that f **vanishes** at $p \in X$ if $f(p) = 0$.

Proposition 2.3.10

If f is a function on a locally ringed space X , then the subset of where f vanishes is closed. Conversely if f vanishes nowhere then it is invertible.

Proof

Note that f does not vanish at p iff $f_p \notin \mathfrak{m}_p$ iff f_p is invertible in $\mathcal{O}_{X,p}$. For a ringed space X the subset of points where f is invertible in the stalk is open. Indeed, if f is invertible in $\mathcal{O}_{X,p}$ then there is a $p \in U$ and a $g \in \mathcal{O}_X(U)$ such that $fg = 1$ in $\mathcal{O}_{X,p}$ so that $fg = 1$ in a neighborhood around p ; so f is invertible in that neighborhood.

Thus the set of points where f is not invertible, for X locally ringed $f(p) = 0$, is closed.

For the second point, at every $p \in X$ there is an open U_p which is the inverse of f in U_p . Because inverses are unique, the g_p agree on their overlaps, so we can glue them together to get a global inverse for f . ■

For an affine scheme $\text{Spec } A$ and a point $[\mathfrak{p}] \in \text{Spec } A$, the stalk at $[\mathfrak{p}]$ is isomorphic to $A_{\mathfrak{p}}$. This is local with maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$, so that $\kappa(\mathfrak{p}) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$. Localization commutes with quotients, so this is isomorphic to the localization of $A/\mathfrak{p}$ at the image of $\mathfrak{p}$, which is the zero ideal. Thus

$$\kappa(\mathfrak{p}) = \frac{A_{\mathfrak{p}}}{\mathfrak{p}A_{\mathfrak{p}}} = \text{Frac}(A/\mathfrak{p}).$$

Example 2.3.11

Consider the scheme $\mathbb{A}_k^2 = \text{Spec } k[x, y]$, where k is a field of characteristic not 2. Then $(x^2 + y^2)/x(y^2 - x^5)$ is a function away from the y -axis and $y^2 - x^5$ (i.e. it exists in $D(y, y^2 - x^5)$). Its value at $(2, 4)$, meaning $(x - 2, y - 4)$ is

$$\frac{x^2 + y^2}{x(y^2 - x^5)} \equiv \frac{2^2 + 4^2}{2(4^2 - 2^5)}.$$

Its value at (y) , the generic point of the x -axis, is

$$\frac{x^2 + y^2}{x(y^2 - x^5)} \equiv \frac{x^2}{-x^6} = -\frac{1}{x^4}.$$

Remark 2.3.12

We can define manifolds of arbitrary flavor similar to schemes. Instead of our base space being $\text{Spec } A$, we can choose different base spaces. For instance a topological manifold is a ringed space $(X, \mathcal{O}_X)$ which is locally isomorphic to an open ball in $\mathbb{R}^n$ with the sheaf of real-valued continuous functions on it. Similarly a C^k -manifold is a ringed space which is locally isomorphic to an open in $\mathbb{R}^n$ with the sheaf of real-valued C^k functions on it. And a complex manifold is a ringed space locally isomorphic to an open ball in $\mathbb{C}^n$ with the sheaf of holomorphic complex-valued functions on it.

In each of these examples we must also require that X be second-countable and Hausdorff.

Theorem 2.3.13

There is a contravariant equivalence of categories $\text{Spec: Ring}^{\text{op}} \rightarrow \text{AffSch: } \Gamma$.

Proof

Because

$$\Gamma \operatorname{Spec} A = \operatorname{Spec} A(D 1) = A_1 = A$$

naturally, it is sufficient to show that Spec is fully faithful (as it is by definition essentially surjective). First we note that for a ring morphism $\varphi: A \rightarrow B$, we can obtain φ from $\operatorname{Spec} \varphi: \operatorname{Spec} B \rightarrow \operatorname{Spec} A$ as $\varphi = \Gamma \operatorname{Spec} \varphi$. Indeed $\operatorname{Spec} \varphi$ acts on global sections by the canonical map $A = A_1 \rightarrow B_{\varphi 1} = B$, which is φ . Thus Spec is faithful.

Similarly if $\varphi: \operatorname{Spec} B \rightarrow \operatorname{Spec} A$ is a morphism of schemes, then we claim $\varphi = \operatorname{Spec} \Gamma \varphi$. Indeed, let $\varphi_1 = \Gamma \varphi: A \rightarrow B$ and $\psi = \operatorname{Spec} \varphi_1$; further let φ denote the underlying topological map of φ and $\varphi^\sharp: \mathcal{O}_{\operatorname{Spec} A} \rightarrow \varphi_* \mathcal{O}_{\operatorname{Spec} B}$ be the sheaf morphism. By definition for $\mathfrak{q} \in \operatorname{Spec} B$, $\varphi_q^\sharp: \mathcal{O}_{\operatorname{Spec} A, \varphi_q} \rightarrow \mathcal{O}_{\operatorname{Spec} B, \mathfrak{q}}$ is local; meaning

$$(\varphi_q^\sharp)^{-1}(\mathfrak{m}_q) = \mathfrak{m}_{\varphi_q}.$$

Recalling $\mathcal{O}_{\operatorname{Spec} A, \mathfrak{p}} \cong A_{\mathfrak{p}}$, we have the following commutative square

$$\begin{array}{ccc} A & \xrightarrow{\varphi_1} & B \\ \downarrow & & \downarrow \\ A_{\varphi_q} & \xrightarrow{\varphi_q^\sharp} & B_{\mathfrak{q}} \end{array}$$

The image of $a \in A$ in the diagonal is then $\varphi_q^\sharp(a/1) = \varphi_1(a)/1$. Thus $\varphi_q^\sharp(a/1) \in \mathfrak{m}_q$ iff $\varphi_1(a) \in \mathfrak{q}$. The former occurs only if $a/1 \in \mathfrak{m}_{\varphi_q}$ iff $a \in \varphi_q$. So $a \in \varphi_q$ iff $\varphi_1(a) \in \mathfrak{q}$, so that $\varphi_1^{-1}\mathfrak{q} = \varphi_q$; meaning $\varphi_1^* = \varphi$ as continuous maps.

Now we need to show that $\varphi_1^\sharp = \varphi^\sharp$ as sheaf maps. We do this on the distinguished open subsets. For $f \in A$ we have that

$$\varphi^{-1} D_A(f) = (\varphi_1^*)^{-1} D_A(f) = \{\mathfrak{q} \leq B \mid f \notin \varphi_1^{-1}\mathfrak{q}\} = D_B(\varphi_1 f).$$

And we want to show that

$$\varphi_f^\sharp: \mathcal{O}_{\operatorname{Spec} A}(D f) = A_f \rightarrow \mathcal{O}_{\operatorname{Spec} B}(\varphi^{-1} D f) = B_{\varphi_1 f}$$

is equal to induced map $A_f \rightarrow B_{\varphi_1 f}$ given by $A \xrightarrow{\varphi_1} B \rightarrow B_{\varphi_1 f}$. But this immediate from the naturality of φ : the following diagram commutes

$$\begin{array}{ccc} A & \xrightarrow{\varphi_1} & B \\ \downarrow & & \downarrow \\ A_f & \xrightarrow{\varphi_f^\sharp} & B_{\varphi_1 f} \end{array}$$

Thus we get the desired result. ■

Example 2.3.14

Let $A = k[x, y]$, so $\operatorname{Spec} A = \mathbb{A}_k^2$; define $U = \mathbb{A}_k^2 - \{(0, 0)\} = \mathbb{A}_k^2 - \{(x, y)\}$. Note that $U = D(x) \cup D(y)$, which forms an affine cover so that U is a scheme. We claim that U is not affine; and thus not a distinguished open set.

Indeed, inclusion $U \hookrightarrow \operatorname{Spec} A$ induces a map $\Gamma U \rightarrow A$. We claim it is an isomorphism: this is sufficient as the inclusion is not an isomorphism, and if U were affine as Γ is an equivalence this would imply it were.

Indeed, functions on U are functions on $D(x)$ and $D(y)$ which agree on their intersection: $D(x) \cap D(y) = D(xy)$. I.e. rational maps in A_x and A_y which are in A_{xy} . These are just the polynomials so $\Gamma U = k[x, y] = A$.

We can extend the result of theorem 2.3.13. The proof is similar, and we rewrite it.

Lemma 2.3.15

The global sections functor $\Gamma: \text{LocRing}^{\text{op}} \rightarrow \text{Ring}$ is fully faithful on morphisms from a locally ringed space to an affine scheme. That is for a locally ringed space X and a ring A ,

$$\text{LocRing}(X, \text{Spec } A) \xrightarrow{\Gamma} \text{Ring}(A, \Gamma X)$$

is bijective.

Proof

Suppose we have two morphisms $\varphi, \psi: X \rightarrow \text{Spec } A$ where X is a locally ringed space, such that $\Gamma\varphi = \Gamma\psi$. Let us write $\varphi^\#$ for the sheaf morphism part of φ , and similar for ψ ; we use φ for the continuous part. Furthermore write φ_1 for $\Gamma\varphi$, similar for ψ . Let $p \in X$: we want to show that $\varphi(p) = \psi(p)$.

Let $\mathfrak{q}_1 = \varphi(p)$ and $\mathfrak{q}_2 = \psi(p)$, and consider the following commutative square:

$$\begin{array}{ccc} A & \xrightarrow{\varphi_1} & \Gamma X \\ \downarrow & & \downarrow \text{loc} \\ A_{\mathfrak{q}_1} & \xrightarrow{\varphi_p} & \mathcal{O}_{X,p} \end{array}$$

Because φ_p is a local morphism we have that for $a \in A$, $a \in \mathfrak{q}_1$ iff $\varphi_p(a/1) \in \mathfrak{m}_{X,p}$. This is iff $\text{loc} \circ \varphi_1(a) \in \mathfrak{m}_{X,p}$. Because $\varphi_1 = \psi_1$, this means that $a \in \mathfrak{q}_1$ iff $a \in \mathfrak{q}_2$ so that $\varphi(p) = \psi(p)$ as desired.

Now we want to show that $\varphi^\# = \psi^\#$; we can do this on distinguished open sets. For $f \in A$ we have the commutative square

$$\begin{array}{ccc} A & \xrightarrow{\varphi_1} & \mathcal{O}_X(X) \\ \downarrow & & \downarrow \\ A_f & \xrightarrow{\varphi_f} & \mathcal{O}_X(\varphi^{-1} D(f)) \end{array}$$

Note that by localization φ_f is the unique morphism making this square commute. Since $\varphi_1 = \psi_1$ this implies $\varphi_f = \psi_f$. Thus Γ is faithful.

For fullness, suppose $\sigma: A \rightarrow \Gamma X$ is a ring morphism. For $p \in X$ let $\sigma_p: A \xrightarrow{\sigma} \Gamma X \rightarrow \mathcal{O}_{X,p}$ be the composite and define $\varphi(p) = \sigma_p^{-1}(\mathfrak{m}_{X,p})$. The preimage of a maximal ideal is prime, so φ forms a map $X \rightarrow \text{Spec } A$. It is continuous: for $f \in A$:

$$\varphi^{-1} D(f) = \{p \in X \mid \sigma_p(f) \notin \mathfrak{m}_{X,p}\} = \{p \in X \mid \sigma(f)(p) \neq 0\},$$

where $\sigma(f)(p)$ is the value of the global section $\sigma(f)$ at p . By proposition 2.3.10 this is an open set.

Now we define $\varphi^\#: \text{Spec } A \rightarrow \varphi_* X$ again on the distinguished open sets. For $f \in A$, $\sigma(f)$ vanishes nowhere in $\varphi^{-1} D(f)$, so it is invertible in $\mathcal{O}_X(\varphi^{-1} D f)$ again by proposition 2.3.10. Thus there is a unique map $\varphi_f: A_f \rightarrow \mathcal{O}_X(\varphi^{-1} D f)$ making the following square commute

$$\begin{array}{ccc} A & \xrightarrow{\sigma} & \mathcal{O}_X(X) \\ \downarrow & & \downarrow \\ A_f & \xrightarrow{\varphi_f} & \mathcal{O}_X(\varphi^{-1} D f) \end{array}$$

This forms a morphism of sheaves, and $(\varphi, \varphi^\#)$ is a morphism of locally ringed spaces. For $p \in X$, let $\mathfrak{q} = \varphi(p) = \sigma_p^{-1}(\mathfrak{m}_{X,p})$. We want to show that $\varphi_p(\mathfrak{q}A_{\mathfrak{q}}) \subseteq \mathfrak{m}_{X,p}$. By considering the following diagram

$$\begin{array}{ccc}
A & \xrightarrow{\sigma} & \mathcal{O}_X(X) \\
\downarrow & & \downarrow \text{loc} \\
A_{\mathfrak{q}} & \xrightarrow{\varphi_p} & \mathcal{O}_{X,p}
\end{array}$$

it is sufficient to show that $\text{loc} \circ \sigma(\mathfrak{q}) \subseteq \mathfrak{m}_{X,p}$. By definition this is just $\sigma_p(\mathfrak{q}) \subseteq \mathfrak{m}_{X,p}$, which is clear by definition of $\mathfrak{q}$. ■

Theorem 2.3.16

The functors $\text{Spec}: \text{Ring}^{\text{op}} \rightarrow \text{LocRing}$ and $\Gamma: \text{LocRing}^{\text{op}} \rightarrow \text{Ring}$ are mutually right adjoint. That is, there is a natural isomorphism

$$\text{LocRing}(X, \text{Spec } A) \cong \text{Ring}(A, \Gamma X), \quad A \in \text{Ring}, X \in \text{LocRing}.$$

We can restrict this adjunction to Sch instead of LocRing .

Proof

By the previous theorem, $\text{LocRing}(X, \text{Spec } A) \xrightarrow{\Gamma} \text{Ring}(A, \Gamma X)$ is a bijection, thus a natural isomorphism. ■

An immediate corollary is the following.

Corollary 2.3.17

The global sections and Spec functors contravariantly preserve colimits.

Note:

By *contravariantly* preserving (co)limits, we mean it maps limiting cones to colimiting cones or colimiting cones to limiting cones.

2.4 Projective schemes

Lemma 2.4.1

Suppose we have a collection of schemes $\{X_i\}_{i \in I}$, open subschemes $X_{ij} \subseteq X_i$ for $i, j \in I$ with $X_{ii} = X_i$, and isomorphisms $f_{ij}: X_{ij} \xrightarrow{\sim} X_{ji}$. Further suppose that these isomorphisms satisfy the **cocycle condition**:

$$f_{ik}|_{X_{ij} \cap X_{ik}} = f_{jk}|_{X_{ji} \cap X_{jk}} \circ f_{ij}|_{X_{ij} \cap X_{ik}}.$$

In this condition we leave implicit that $f_{ij}(X_{ij} \cap X_{ik}) \subseteq X_{jk}$.

Then there exists a unique scheme X which has open subschemes isomorphic to X_i whose intersections are isomorphic to X_{ij} , which respects the gluing data in the obvious sense.

Proof

TODO: This

Example 2.4.2 (Projective space)

Let A be a ring, we define **projective n -space** $\mathbb{P}_A^n$ by gluing together $n+1$ copies of $\mathbb{A}_A^n = \text{Spec } A[x_1, \dots, x_n]$ in a particular manner. Let us denote these copies by $U_0, \dots, U_n$ where we write U_i as

$$U_i = \text{Spec } A[x_{0/i}, \dots, x_{n/i}]/(x_{i/i} - 1).$$

Then we let $U_{ij} = D(x_{j/i})$, and we identify U_{ij} and U_{ji} via the isomorphism

$$A[x_{0/i}, \dots, x_{n/i}, 1/x_{j/i}]/(x_{i/i} - 1) \cong A[x_{0/j}, \dots, x_{n/j}, 1/x_{i/j}]/(x_{j/j} - 1)$$

which maps $x_{k/i}$ to $x_{k/j}/x_{i/j}$. Because Spec is an equivalence, this forms an isomorphism of schemes.

It is easy to see that these isomorphisms satisfy the cocycle conditions, so we can glue them together to get the scheme $\mathbb{P}_A^n$.

We can write $\mathbb{P}_A^n$ as a colimit of the U_i and U_{ij} s, and because Γ contravariantly preserves colimits (corollary 2.3.17), we can write $\Gamma(\mathbb{P}_A^n)$ as a limit of $A[x_{0/i}, \dots, x_{n/i}]/(x_{i/i} - 1)$ and $A[x_{0/i}, \dots, x_{n/i}, 1/x_{j/i}]/(x_{i/i} - 1)$. It is easy to see that this colimit is isomorphic to A , so that $\Gamma(\mathbb{P}_A^n) \cong A$. But for $n > 0$, $\mathbb{P}_A^n$ is not isomorphic to $\text{Spec } A$, so $\mathbb{P}_A^n$ is not affine.

We can generalize the previous example.

Definition 2.4.3

If M is a commutative monoid, a **M -graded ring** is a ring $S_\bullet = \bigoplus_{m \in M} S_m$ (where S_m are Abelian groups, whose elements are called **homogeneous of degree m**) such that multiplication in $S_\bullet$ respects the grading: it forms maps $S_m \times S_n \rightarrow S_{n+m}$ for $n, m \in M$.

Clearly then S_0 is a subring and $S_\bullet$ is an S_0 -algebra. An ideal I of $S_\bullet$ is **homogeneous** if it is generated by homogeneous elements.

If $S_\bullet$ is M -graded, then there are obvious projections $S_\bullet \rightarrow S_m$ for all $m \in M$; the image of $a \in S_\bullet$ under this projection is called its **degree m part** and is denoted a_m . An ideal $I \leq S_\bullet$ is homogeneous iff it is closed under these projections: if $a \in I$ then $a_m \in I$. Clearly if this holds then I is homogeneous: it is generated by the images of itself under all the projections. Conversely if I is homogeneous, say generated by I_{hom} , then it is easily seen that any combination of elements in I_{hom} have degree m parts which are also combinations of I_{hom} .

Thus a homogeneous ideal can be decomposed into homogeneous pieces $I = \bigoplus_{m \in M} I_m$ where $I_m \leq S_m$ is a subgroup (and $I_0 \leq S_0$ is an ideal). Thus $S_\bullet/I \cong \bigoplus_{m \in M} S_m/I_m$ has a natural M -grading.

Clearly the sum and product of homogeneous ideals are homogeneous. The intersection of homogeneous ideals is homogeneous as well: if $a \in I \cap J$ then $a_m \in I \cap J$ for all $m \in M$ so by the above consideration $I \cap J$ is homogeneous. And if I is homogeneous, so is $\sqrt{I}$: if $a^n \in I$ then

$$(a^n)_m = \sum_{k_1 + \dots + k_n = m} a_{k_1} \cdots a_{k_n}$$

is in I as all the a_{k_i} are.

If M is a commutative monoid, we associate to it the Abelian group $\widetilde{M}$ which is the quotient of the monoid $M \times M$ by the equivalence $(m, n) \sim (m + k, n + k)$ for all $k \in M$. We can embed $M \rightarrow \widetilde{M}$ by $m \mapsto (m, 0)$, and we note that this is the identity of an adjunction $\text{CommMon} \nrightarrow \text{Ab}$.

If $T \subseteq S_\bullet$ is a multiplicative subset containing only homogeneous elements, then $T^{-1}S_\bullet$ has an obvious grading over $\widetilde{M}$ where a_m/t_n has degree $m - n$. It is easy to check that this is a well-defined grading.

Note:

In order to avoid excessive subscripts, we will also use the notation $S_\bullet[T^{-1}]$ for $T^{-1}S_\bullet$. In particular, $(S_\bullet)_f$ will instead be denoted $S_\bullet[1/f]$.

Now we consider only the case where $M = \mathbb{Z}$ or $\mathbb{Z}_{\geq 0}$ (every $\mathbb{Z}_{\geq 0}$ -grading can be turned into a $\mathbb{Z}$ grading by setting $S_n = 0$ for $n < 0$). We define the **degree** of $a \in S_\bullet$ to be

$$\deg(a) = \max \{n \in \mathbb{Z} \mid a_n \neq 0\}.$$

In order for an homogeneous ideal $I < S_\bullet$ to be prime, it is sufficient to check the usual condition on homogeneous elements only: that is, if $ab \in I$ then $a, b \in I$ for homogeneous a, b . Suppose that $a, b \notin I$, then define $a_I = \sum_{m, a_m \in I} a_m$ and $a^I = \sum_{m, a_m \notin I} a_m$ so that $a = a_I + a^I$, thus

$$ab = a_I b_I + a_I b^I + a^I b_I + a^I b^I.$$

As $a_I \in I$ we have that $a^I \notin I$, and we need to show that $a^I b^I \notin I$. That is, it is sufficient to consider the case where $a_m \notin I$ for all $m \in M$ (or is zero). Let $n = \deg(a)$ and $m = \deg(b)$ so that $(ab)_{n+m} = a_n b_m$. Because $a_n, b_m \notin I$ we have that $(ab)_{n+m} \notin I$ (by primality on homogeneous elements) and thus $ab \notin I$ (because I is homogeneous).

Note:

For the remainder of these notes, $S_\bullet$ will be a $\mathbb{Z}_{\geq 0}$ -graded ring unless otherwise specified.

If $S_0 = A$, then $S_\bullet$ is said to be **graded over** A . For example $A[x_0, \dots, x_n]$ is graded over A (where the homogeneous elements are homogeneous polynomials), or $A[x_0, \dots, x_n]/I$ where I is an homogeneous ideal and $I \cap A = 0$. For a graded ring $S_\bullet$, $S_+ = \bigoplus_{n>0} S_n$ is an ideal called the **irrelevant ideal**. If S_+ is a finitely generated ideal, then $S_\bullet$ is said to be **finitely generated over** A .

Note that $S_\bullet$ is finitely generated over A iff $S_\bullet$ is a finitely generated graded A -algebra (i.e. generated by finitely many homogeneous elements). If $S_\bullet$ is finitely generated over A , so $S_+ = (f_1, \dots, f_n)$ homogeneous, then by considering degrees we see that S_1 is A -spanned by the f_i . Then we can see that S_2 is A -spanned by the $f_i, f_i f_j$ s, and so on by induction. That is, S_k is spanned by the products of f_i s, so that the f_i s generate $S_\bullet$ as an A -algebra. Conversely if $S_\bullet$ is finitely generated as a graded A -algebra by $f_1, \dots, f_n$, then these must also generate S_+ .

Furthermore $S_\bullet$ is Noetherian iff A is, and $S_\bullet$ is finitely generated over A . Indeed if $S_\bullet$ is Noetherian, then S_+ is finitely generated as it is an ideal; by the previous paragraph this means that $S_\bullet$ is finitely generated. And any ideal of A can be extended to an homogeneous ideal of $S_\bullet$ (with the same generators), which is finitely generated. It is clear that the degree zero generators of this ideal generate the ideal from A . Alternatively we could note that the extension of A 's ideal is unique, as it adds no additional elements from A , and consider the definition of a Noetherian ring. Conversely if $S_\bullet$ is finitely generated and A is Noetherian, then $S_\bullet$ is a Noetherian A -algebra and thus a Noetherian ring.

Now we can see why graded rings are important. The idea is that for a graded ring $S_\bullet$ we will define a scheme $\text{Proj } S_\bullet$, motivated by $\mathbb{P}_A^n$. That is, $\text{Proj } A[x_0, \dots, x_n]$ will give us $\mathbb{P}_A^n$.

Proposition 2.4.4

Let $f \in S_+$ be homogeneous. Then there is a bijection between the prime ideals of $(S_\bullet)[1/f]_0$ and the homogeneous prime ideals of $(S_\bullet)[1/f]$.

Proof

We generalize this result as follows: for any $\mathbb{Z}$ -graded ring A with an invertible homogeneous element $f \in A$ of positive degree, there is a bijection between the prime ideals of A_0 and the homogeneous prime ideals of A .

Consider the ring map $A_0 \rightarrow A$, which then maps homogeneous prime ideals of A to prime ideals of A_0 via the inverse image. Note that this maps a prime ideal $P = \bigoplus_i Q_i \leq A$ to Q_0 . Conversely if $P_0 \leq A_0$ is a prime ideal, then define $P \leq A$ as $P = \bigoplus_i Q_i$ where $a \in Q_i$ iff $a^{\deg f}/f^i \in P_0$. Here we have that $Q_0 = P_0$. It remains to show that P is an ideal, and prime.

Note that $a \in Q_i$ iff $a^2 \in Q_{2i}$; $a^{\deg f}/f^i \in P_0$ iff $a^{2\deg f}/f^{2i} \in P_0$ as P_0 is prime, iff $a^2 \in Q_{2i}$. And if $a_1, a_2 \in Q_i$ then $(a_1 + a_2)^2 \in Q_{2i}$; indeed by the binomial theorem (letting $d = \deg f$)

$$\frac{(a_1 + a_2)^{2d}}{f^{2i}} = \frac{\sum_k \binom{2d}{k} a_1^k a_2^{2d-k}}{f^{2i}},$$

and for all k , one of k or $2d - k$ is at least d , so each summand is in P_0 . Thus $a_1 + a_2 \in Q_i$, meaning P is closed under addition.

And if $a \in Q_i$ and $b \in S_\bullet$ has degree k , then $ab \in Q_{i+k}$ because

$$\frac{(ab)^d}{f^{i+k}} = \frac{a^d}{f^i} \cdot \frac{b^d}{f^k} \in P_0.$$

Thus P is closed under multiplication so it is an ideal.

It is sufficient to check primality on homogeneous elements, so if $a, b \in S_\bullet$ are homogeneous of degrees i, j respectively, then if $ab \in Q_{i+j}$ that means $(ab)^d/f^{i+j} \in P_0$. By primality $a^d/f^i, b^d/f^j \in P_0$ so $a \in Q_i$ and $b \in Q_j$ as desired. Thus $P \leq S_\bullet$ is a prime ideal.

Now if $P \leq S_\bullet$ is a prime ideal with $P = \bigoplus_i P_i$, we want to show that $a \in P_i$ iff $a^d/f^i \in P_0$. Indeed if $a^d/f^i \in P_0$ then by primality $a \in P_i$. Conversely if $a \in P_i$ then $a^d/f^i \in P$ is homogeneous of degree 0, so it is in P_0 as desired. So the two maps are mutual inverses, as desired. ■

Now we can construct $\text{Proj } S_\bullet$ as the gluing of $\text{Spec } S_\bullet[1/f]_0$ along the $U_{fg} = D(g^{\deg f}/f^{\deg g})$, where f ranges over the homogeneous elements of positive degree. Note that for $S_\bullet = A[x_0, \dots, x_n]$ and $f = x_i$ then $S_\bullet[1/f]_0 \cong A[x_{0/i}, \dots, x_{n/i}]/(x_{i/i} - 1)$ by $x_{k/i} \mapsto x_k/x_i$. And $U_{ij} = D(x_j/x_i)$ corresponds to $D(x_{j/i})$, so this gives us the original definition of $\mathbb{P}_A^n$ (we will see that we need only glue along generators).

We note that U_{fg} is isomorphic to $\text{Spec } S_\bullet[1/f]_0$. This isomorphism is induced from the map $S_\bullet[1/f] \rightarrow S_\bullet[1/fg]$. Thus there is an isomorphism $U_{fg} \cong U_{gf}$, and it is routine to check that these isomorphisms satisfy the cocycle condition. Thus we have defined $\text{Proj } S_\bullet$; we will continue this section by providing more descriptions of $\text{Proj } S_\bullet$.

By the above proposition, the prime ideals of $S_\bullet[1/f]_0$ correspond to the homogeneous prime ideals of $S_\bullet[1/f]$. We can construct a bijection between the points of $\text{Proj } S_\bullet$ and the homogeneous prime ideals of $S_\bullet$. For a homogeneous prime ideal $\mathfrak{p} \leq S_\bullet$, then for all $f \notin \mathfrak{p}$ we have the homogeneous prime ideal $\mathfrak{p}_f \in S_\bullet[1/f]$. All of these ideals are identified: if $f, g \notin \mathfrak{p}$ then $fg \notin \mathfrak{p}$ so that $\mathfrak{p}_f$ and $\mathfrak{p}_g$ are identified in $\text{Spec } S_\bullet[1/fg]_0$. And all of homogeneous prime ideals of $S_\bullet[1/f]$ are of this form, so mapping $\mathfrak{p} \mapsto \mathfrak{p}_f$ is surjective. Because $\mathfrak{p} \mapsto \mathfrak{p}_f$ is an injection (for fixed f , $f \notin \mathfrak{p}$), we see that distinct ideals are not identified.

Note that we must consider only $\mathfrak{p}$ which don't contain some homogeneous $f \in S_+$; that is S_+ cannot be a subset of $\mathfrak{p}$. This is because we consider only f with positive degree. These homogeneous prime ideals which do not contain S_+ are called **relevant**; otherwise they are **irrelevant**.

Thus the points of $\text{Proj } S_\bullet$ correspond to the relevant homogeneous prime ideals of $S_\bullet$.

Definition 2.4.5

Let $T \subseteq S_\bullet$ be a collection of homogeneous elements. Define its **vanishing set** to be

$$V(T) = \{\mathfrak{p} \leq S_\bullet \text{ homogeneous prime} \mid T \subseteq \mathfrak{p}, \text{ and } \mathfrak{p} \text{ does not contain the irrelevant ideal}\}.$$

And if f is a homogeneous element of *positive degree*, define $D_+(f) = \text{Proj } S_\bullet - V(f)$; its **projective distinguished open set**.

Note that $D_+(f)$ can be written as

$$D_+(f) = \{\mathfrak{p} \leq S_\bullet \text{ homogeneous prime} \mid f \notin \mathfrak{p}\}.$$

It is immediat that this does not contain any irrelevant ideals as $f \in S_+$. Furthermore this corresponds to $\text{Spec } S_\bullet[1/f]_0$. Indeed, the homogeneous prime ideals of $S_\bullet[1/f]$ correspond to the homogeneous prime ideals of $S_\bullet$ not containing f .

The projective vanishing sets satisfy the same properties as those in the affine case. Thus they form a topology of closed sets on $\text{Proj } S_\bullet$, and $D_+(f)$ form a basis.

Note that any homogeneous ideal $I \leq S_\bullet$ is contained in a homogeneous prime ideal. Indeed, for an ideal I define its **homogenization** to be

$$I_{\text{hom}} = \bigoplus_n (I \cap S_n) = (a \in I \mid a \text{ is homogeneous}).$$

Because primality can be checked on homogeneous elements, if $\mathfrak{p}$ is prime so is $\mathfrak{p}_{\text{hom}}$. Thus if $I \leq \mathfrak{p}$ is prime, then $I \leq \mathfrak{p}_{\text{hom}}$ is homogeneous prime. Note that because $\mathfrak{p}_{\text{hom}} \leq \mathfrak{p}$, we have that

$$\sqrt{I} = \bigcap_{I \leq \mathfrak{p} \text{ homogeneous}} \mathfrak{p}.$$

Here the intersection is almost over $V(I)$, but it may be the case that the $\mathfrak{p}$ s are irrelevant. We will remedy this now.

Note that the following are equivalent for any homogeneous ideal $I \subseteq S_+$.

- (1) $V(I) = \emptyset$,
- (2) if $I = (f_i)_i$ are homogeneous generators, then $\bigcup_i D_+(f_i) = \text{Proj } S_\bullet$,
- (3) $\sqrt{I} \supseteq S_+$.

(1) and (2) are equivalent because

$$V(f_i)_i = \bigcap_i V(f_i),$$

which is then empty iff its complement is $\text{Proj } S_\bullet$, which is precisely (2). Note that $V(I) = \emptyset$ iff all of the homogeneous prime ideals containing I also contain S_+ . This is iff their intersection contains S_+ . Because this intersection is $\sqrt{I}$, this gives us (3).

Call an ideal I satisfying any of the above equivalent conditions **irrelevant**. Clearly by (3) this agrees with our definition of an irrelevant prime ideal.

For a subset $Z \subseteq \text{Proj } S_\bullet$, define the relevant homogeneous ideal $I(Z) \leq S_\bullet$ by

$$I(Z) = \bigcap_{\mathfrak{p} \in Z} \mathfrak{p}.$$

The intersection of homogeneous ideals is homogeneous, so $I(Z)$ is homogeneous. Then we have that

$$V I(Z) = \overline{Z}$$

whose proof is the same as in the affine case.

We also have that for $I \leq S_+$ relevant homogeneous,

$$I V(I) = \sqrt{I}.$$

Indeed because $I \leq S_+$ we can ignore degree zero, in which case all irrelevant prime ideals all look like S_+ , and thus contribute nothing to the intersection, as there exists a relevant prime ideal. Thus the intersection of all homogeneous prime ideals containing I looks the same as the intersection of all relevant prime ideals containing I .

Thus as in the affine case, this clearly demonstrates that $V(I) \subseteq V(J)$ iff $\sqrt{J} \leq \sqrt{I}$.

Now we want to define the structure sheaf on $\text{Proj } S_\bullet$. Recall that there is a correspondence $\varphi: D_+(f) \cong \text{Spec } S_\bullet[1/f]_0$; we claim that this correspondence is a homeomorphism. Indeed, the map $\mathfrak{p} \mapsto (\mathfrak{p}_f)_0$ is continuous: for $g/f^k \in S_\bullet[1/f]_0$ we have that

$$\varphi^{-1} D(g/f^k) = \{\mathfrak{p} \in D_+(f) \mid g/f^k \notin (\mathfrak{p}_f)_0\}.$$

This is equal to $\{\mathfrak{p} \mid g \notin \mathfrak{p}\}$; because $f \notin \mathfrak{p}$, we have that $g/f^k \in \mathfrak{p}_f$ iff $gf^t \in \mathfrak{p}$ for some t iff $g \in \mathfrak{p}$. Thus

$$\varphi^{-1} D(g/f^k) = D_+(g)$$

so that φ is continuous. And it is open, since for g homogeneous

$$\varphi D_+(fg) = \{(\mathfrak{p}_f)_0 \mid fg \notin \mathfrak{p}\} = D\left(\frac{g^{\deg f}}{f^{\deg g}}\right).$$

The last equality is because $g^d/f^k \notin \mathfrak{p}_f$ iff $g^d \notin \mathfrak{p}$ iff $g \notin \mathfrak{p}$ when $f \notin \mathfrak{p}$.

Now $D_+(fg)$ corresponds to $D(g^{\deg f}/f^{\deg g})$ in $\text{Spec } S_\bullet[1/f]_0$ and to $D(f^{\deg g}/g^{\deg f})$ in $\text{Spec } S_\bullet[1/g]_0$. We want to define the structure sheaf on $D_+(f) \subseteq \text{Proj } S_\bullet$ to be the structure sheaf of $\text{Spec } S_\bullet[1/f]_0$. It is routine to check that this agrees with triple intersections (the cocycle condition).

Note that if $S_+ = (f_i)_i$ then we can consider only the $D_+(f_i)$. Indeed, this comes down to the fact that if $X = \bigcup_i U_i$ then any sheaf $(X, \mathcal{O}_X)$ can be written as the gluing of the $(U_i, \mathcal{O}_{U_i})$.

So in terms of $A[x_0, \dots, x_n]$, we have that the x_i generate the irrelevant ideal, so we can focus on the structure sheaf restricted to

$$D_+(x_i) \cong \text{Spec } A[x_0, \dots, x_n, 1/x_i]_0 \cong \text{Spec } A[x_{0/i}, \dots, x_{n/i}]/(x_{i/i} - 1).$$

Thus defining $\mathbb{P}_A^n = \text{Spec } A[x_0, \dots, x_n]$ agrees with our previous definition.

2.5 Properties of schemes

2.5.1 Topological properties

Proposition 2.5.1

If X is a topological space with a finite open cover $\{X_i\}_{i=1}^n$ such that each X_i is irreducible and $\bigcap_i X_i \neq \emptyset$, then X is irreducible.

Proof

Let $Y = \bigcap_i X_i$; as a finite intersection of open sets it is open. For $U, V \subseteq X$ nonempty open, we must have that $U \cap X_i \neq \emptyset$ for some i . Since X_i is irreducible and $Y \subseteq X_i$ is open, we must have that $U \cap Y$ is nonempty. Similarly $V \cap Y$ is nonempty, and these are both nonempty open subsets of any X_i , so they have nonempty intersection. Thus $U \cap V \neq \emptyset$ as desired. ■

Corollary 2.5.2

If A is a domain, then $\mathbb{P}_A^n$ is irreducible.

Proof

We know that $D_+(x_i) \cong \text{Spec } A[x_0, \dots, x_n]$ is irreducible by corollary 2.1.8. And the intersection of all these $D_+(x_i)$ is nonempty (it contains the zero ideal). ■

Recall that for an affine scheme, there is a bijection between points and irreducible closed subsets, given by $p \mapsto \overline{\{p\}}$. Recall this is because closed irreducible subsets are all of the form $V(\mathfrak{p})$ for a prime ideal $\mathfrak{p}$. This is true for schemes in general. Indeed, if $p \mapsto \{p\}$ is a bijection between points and closed irreducible subsets of $X_i \subseteq X$ for an open cover $X = \bigcup_i X_i$, then it is one for X as well.

Let $Z \subseteq X$ be closed irreducible, so there is an i such that $Z \cap X_i \subseteq X_i$ is nonempty. Thus there is $p \in X_i$ such that $Z \cap X_i = \text{cl}_{X_i}(p)$. Because $Z \cap X_i \subseteq Z$ is open, it is dense, so that

$$\text{cl}_X(p) = \text{cl}_X(\text{cl}_{X_i}(p)) = \text{cl}_X(Z \cap X_i) = Z$$

as desired.

Now if $\text{cl}(p) = \text{cl}(q) = Z$ then $p = q$. Indeed, because $\{p\}, \{q\}$ are dense in Z , they intersect all of its nonempty open subsets. So choose i such that $Z \cap X_i \neq \emptyset$, then $p, q \in X_i$ and $\text{cl}_{X_i}(p) = \text{cl}_{X_i}(q) = Z \cap X_i$ and then by assumption $p = q$.

Corollary 2.5.3

If X is a scheme, $p \mapsto \{p\}$ is a bijection between points of X and its closed irreducible subsets.

Proof

X has an open cover of affine schemes, which have this property. By the above discussion, X has this property. ■

Note:

A space with this property is called a **Sober space**.

Proposition 2.5.4

If $X = \bigcup_i X_i$ is a finite covering of X by Noetherian subspaces, then X is Noetherian.

Proof

Let $Z_0 \supset Z_1 \supset \dots$ be a descending chain of closed subsets of X . Then intersecting with X_i gives a stabilizing chain, say it stabilizes at n_i . Then the original chain stabilizes at $N = \max n_i$ (because the original chain is the union of the intersection with the X_i s). ■

Corollary 2.5.5

If X is a scheme with a finite cover $X = \bigcup_{i=1}^n \text{Spec } A_i$ where each A_i is Noetherian, then X is a Noetherian space.

Corollary 2.5.6

If A is Noetherian, then $\mathbb{P}_A^n$ is.

Proof

We know that $A[x_0, \dots, x_n]$ is Noetherian by Hilbert's basis theorem, and $\mathbb{P}_A^n$ has a finite covering of the spectrum of this algebra. ■

Note that if X has a finite covering by quasicompact open subsets, then X is quasicompact. This is pretty immediate by taking an open cover and then considering it over each open set. Thus the following is immediate (as affine schemes are quasicompact).

Corollary 2.5.7

A scheme is quasicompact iff it has a finite open affine covering.

Proposition 2.5.8

Let X be a quasicompact scheme, then every closed subset of X has a closed point.

Proof

For affine schemes this is true, because every ideal $I \leq A$ is contained in a maximal ideal, which is a closed point in $V(I)$.

Now let $X = \bigcup_{i=1}^n U_i$ form an affine cover, and let $p_1 \in Z \cap U_1$ be a closed point. If p_1 is closed in X we are done, otherwise $\overline{p_1}$ contains points other than p_1 , which necessarily cannot lie in U_1 . So $\overline{p_1}$ intersects some other U_i , call it U_2 , and then again it contains some closed point (of U_2) $p_2 \in \overline{p_1} \cap U_2$. If p_2 is closed in X we are done, otherwise $\overline{p_2} \subseteq \overline{p_1}$, and by corollary 2.5.3 this is a strict inclusion. Thus it intersects some other U_3 , and we can inductively choose a closed point $p_{i+1} \in \overline{p_i} \cap U_i$.

This process must halt eventually, as there are only n affine sets, so eventually we must find a closed point. ■

Remark 2.5.9

This proposition can help us prove properties of schemes. Specifically suppose we have some property P of schemes which is **open**: if a point p has P , then there is a neighborhood of p , all of whose points have P . Then in order to check if all points of a quasicompact scheme have P , it is sufficient to check that the closed points have P .

Note that $p \in \overline{q}$ iff $N(p) \subseteq N(q)$ (where $N(p)$ is the collection of p 's neighborhoods). Indeed if $p \notin \overline{q}$ then $X - \overline{q}$ is a neighborhood of p which does not contain q . Conversely if $p \in U$ and $q \notin U$, then $\overline{q} \subseteq X - U$ and so $p \notin \overline{q}$.

So if P is open and all closed points of X have P , then let $q \in X$. Choose a closed point $p \in \overline{q}$ so that there is a neighborhood $p \in U$ such that all points of U have P . Because $N(p) \subseteq N(q)$, we have that $q \in U$ and thus q has P .

Definition 2.5.10

A space X is **quasiseparated** if the intersection of any two quasicompact open subsets is quasicompact.

A scheme X is quasiseparated iff the intersection of any two open affine subschemes is a finite union of affine open subsets. If X is quasiseparated this is clear, as affine subschemes are quasicompact, and open subsets are themselves schemes (so to be quasicompact they must have a finite affine covering). Conversely if $U, V \subseteq X$ are quasicompact open, then they each have a finite affine covering, and their intersection then has a finite affine covering, so is quasicompact.

It is sufficient to only consider a basis of affine subschemes, and as such it is easy to see that affine schemes are quasiseparated: $D(f) \cap D(g) = D(fg)$.

Thus if X is covered by affine subschemes, such that the intersection of any two of these affine subschemes is quasicompact. Indeed, if $X = \bigcup_i U_i$ and $U_i \cap U_j = \bigcup_k V_{ijk}$ is a finite union of affine subschemes, then let D_{il} form an affine basis for U_i . Then $D_{il} \cap D_{j\ell'} \subseteq U_i \cap U_j = \bigcup_k V_{ijk}$, and so $D_{il} \cap D_{j\ell'} \cap V_{ijk}$ is quasicompact as V_{ijk} is affine and thus quasiseparated. Thus $D_{il} \cap D_{j\ell'} \cap V_{ijk}$ can be written as a finite union of affine schemes, and taking their unions, so too can $D_{il} \cap D_{j\ell'}$. Because the D_{il} form an affine basis of X , we have the desired result.

It is clear that a scheme X is quasicompact and quasiseparated iff it can be covered by a finite union of affine subschemes which have quasicompact intersection. Quasicompact and quasiseparated is often abbreviated as **qcqs**.

Example 2.5.11

A **projective A -scheme** is a scheme of the form $\text{Proj } S_\bullet$ where $S_0 = A$ and $S_\bullet$ is finitely generated over A . A projective A -scheme is qcqs.

Indeed, let $S_\bullet$ be generated by the homogeneous elements $(x_0, \dots, x_n)$ over A . Then

$$\text{Proj } S_\bullet = \bigcup_{i=0}^n D_+(x_i)$$

is a finite covering of affine subschemes (recall $D(x_i) = \text{Spec } S_\bullet[1/x_i]_0$). Furthermore $D_+(x_i) \cap D_+(x_j) = D_+(x_i x_j)$ is affine as well, showing that $\text{Proj } S_\bullet$ is qcqs as desired.

Example 2.5.12

Let $X = \text{Spec } k[x_1, x_2, \dots]$ and $U = X - [\mathfrak{m}]$ where $\mathfrak{m}$ is the maximal ideal $(x_1, x_2, \dots)$. Then let Y be the scheme obtained by gluing two copies of X along their common copy of U . Now both copies of X in Y are quasicompact (as they are affine), but their intersection is U which is not quasicompact.

Indeed we can cover U as follows

$$U = D(\mathfrak{m}) = \bigcup_{i=1}^{\infty} D(x_i).$$

And any finite union of $D(x_i)$ s is not equal to $D(\mathfrak{m})$ (say because $x_{n+1} \notin \sqrt{(x_1, \dots, x_n)}$).

2.5.2 Reducedness and integrality

Recall a ring A is **reduced** if $\text{nil}(A) = 0$, i.e. it has no nonzero nilpotent elements.

Definition 2.5.13

A scheme is **reduced** if its stalks are all reduced.

Proposition 2.5.14

If f, g are two functions on a reduced scheme which agree at all points, then $f = g$.

Proof

It is sufficient to show that if f is zero at all points, then $f = 0$. That is if $f_p \in \mathfrak{m}_p$ for all $p \in X$, then $f = 0$.

It is sufficient to consider this locally, so we first show this for $X = \text{Spec } A$. We have that $f_p \in \mathfrak{p}A_p$, meaning $f \in \mathfrak{p}$ for all prime $\mathfrak{p} \in \text{Spec } A$. Thus $f \in \text{nil}(A)$, and because ring morphisms preserve nilpotents ($\varphi(f)^n = \varphi(f^n) = 0$), we have that $f_p \in \text{nil}(A_p) = 0$. Thus $f_p = 0$ in A_p for all $\mathfrak{p}$. Because $A \rightarrow \prod_{\mathfrak{p}} A_p$ is injective, this means that $f = 0$.

Now if $X = \bigcup_i \text{Spec } A_i$, then by above we have $f = 0$ in each A_i . By gluing this means $f = 0$ over X . ■

Proposition 2.5.15

A ring A is reduced iff the affine scheme $\text{Spec } A$ is. And a scheme X is reduced iff $\mathcal{O}_X(U)$ is reduced for all open $U \subseteq X$. It is sufficient to check this for an affine open cover $X = \bigcup_i U_i$.

Proof

The first claim amounts to saying A is reduced iff $A_{\mathfrak{p}}$ is for all prime $\mathfrak{p}$. By the injection $A \rightarrow \prod_{\mathfrak{p}} A_{\mathfrak{p}}$, if the $A_{\mathfrak{p}}$ are all reduced then so is A . And if A is reduced, recall that radicals commute with localization, so

$$\text{nil}(A_{\mathfrak{p}}) = \sqrt{0_{\mathfrak{p}}} = \sqrt{0_p} = \text{nil}(A)_{\mathfrak{p}} = 0_{\mathfrak{p}}.$$

Thus $A_{\mathfrak{p}}$ is reduced as desired.

Because being reduced is local, if X is reduced then every open $U \subseteq X$ is reduced. By the injection $\mathcal{O}_X(U) \rightarrow \prod_{p \in U} \mathcal{O}_{X,p}$, we have that $\mathcal{O}_X(U)$ is reduced.

Conversely if there is an affine open cover $X = \bigcup_i U_i$ where $\mathcal{O}_X(U_i)$ are reduced, then the U_i are all reduced as they are isomorphic to $\text{Spec } \mathcal{O}_X(U_i)$, and this is reduced by the first paragraph. Stalks in X can be taken local to a U_i , so we have the desired result. ■

Example 2.5.16

$\mathbb{A}_A^n$ is reduced for any domain A , as $A[x_1, \dots, x_n]$ is a domain and thus reduced. Similarly $\mathbb{P}_A^n$ is reduced for any domain A , because it has an affine covering by $A[x_0, \dots, x_n]$ which are reduced.

Remark 2.5.17

Suppose X is a quasicompact scheme and f vanishes at all points of X . Then there is an n such that $f^n = 0$ in X .

Indeed suppose $X = \bigcup_i \text{Spec } A_i$, following the proof of proposition 2.5.14, we see that $f \in \text{nil}(A_i)$ for all i . So there are n_i such that $f^{n_i} = 0$ in A_i . Taking N to be the maximum n_i , we see that $f^N = 0$ in all A_i , so by gluing $f = 0$.

Definition 2.5.18

A nonempty scheme X is **integral** if $\mathcal{O}_X(U)$ is an integral domain for all open $U \subseteq X$.

Proposition 2.5.19

A scheme is integral iff it is irreducible and reduced.

Proof

If X is integral, then let $U, V \subseteq X$ be nonempty open. If $U \cap V = \emptyset$ then $\mathcal{O}_X(U \sqcup V) = \mathcal{O}_X(U) \times \mathcal{O}_X(V)$ is not a domain. Thus $U \cap V$ is nonempty so X is irreducible. And X is reduced because $\mathcal{O}_X(U)$ is a domain and thus reduced for all U , and by proposition 2.5.15.

Conversely suppose X is irreducible and reduced. Let $U \subseteq X$ be nonempty open, and suppose $f, g \neq 0$ in $\mathcal{O}_X(U)$. Let $U_f = \{p \in U \mid f(p) \neq 0\}$, so that U_f is nonempty by proposition 2.5.14. Because X is irreducible $V = U_f \cap U_g \neq \emptyset$. For $p \in V$ we have that $f(p), g(p) \neq 0$ and thus $(fg)(p) \neq 0$. This means that $fg \neq 0$ in V and thus is nonzero in U as well. ■

Proposition 2.5.20

$\text{Spec } A$ is integral iff A is an integral domain.

Proof

Immediately by definition if $\text{Spec } A$ is integral, then A is a domain.

Conversely if A is an integral domain, it is reduced. Furthermore $\text{nil } A = 0$ is a prime ideal, so A is irreducible by proposition 2.1.7. By the previous proposition this means $\text{Spec } A$ is integral. ■

If X is an irreducible scheme, it is closed and so by proposition 2.5.3 it has a unique generic point $\eta \in X$. Now further suppose that X is integral, and let $\text{Spec } A \subseteq X$ be an affine neighborhood. Because η is dense in X , it is in $\text{Spec } A$ and it must furthermore be a generic point there. Thus $\eta = \text{nil}(A) = 0$ in $\text{Spec } A$, and thus because localization is local

$$\mathcal{O}_{X,\eta} = A_\eta = A_0 = K(A),$$

the fraction field of A .

We denote $\mathcal{O}_{X,\eta}$ by $K(X)$ and call it the **function field** of X ; its elements are **rational functions**.

Note that the canonical map $\mathcal{O}_X(U) \rightarrow K(X) = \mathcal{O}_{X,\eta}$ is injective. Indeed, if $f_\eta = 0$ then $\eta \in \{f_p = 0\}$ which is closed, and since η is generic, f is zero at all points of X . By proposition 2.5.14 this means $f = 0$.

Now if $U \subseteq V$ are nonempty, then inclusion forms a morphism of schemes $U \hookrightarrow V$. Thus localizing at η forms a map $\mathcal{O}_{V,\eta} \rightarrow \mathcal{O}_{U,\eta}$ such that the following diagram commutes

$$\begin{array}{ccc} \mathcal{O}_X(V) & \xrightarrow{\text{res}_{V,U}} & \mathcal{O}_X(U) \\ \downarrow & & \downarrow \\ \mathcal{O}_{V,\eta} & \longrightarrow & \mathcal{O}_{U,\eta} \\ \parallel & & \parallel \\ K(X) & \longrightarrow & K(X) \end{array}$$

Because $K(X)$ is a field, the bottom horizontal arrow is injective. Since the vertical arrows are injective, this means $\text{res}_{V,U}$ is injective as well. Thus we have shown the following.

Proposition 2.5.21

Let X be an integral scheme. Then the restriction maps are all injections, as well as the canonical maps $\mathcal{O}_X(U) \rightarrow K(X)$.

2.5.3 Affine-locality

Proposition 2.5.22

Suppose $\text{Spec } A, \text{Spec } B$ are affine neighborhoods of a scheme X . Then $\text{Spec } A \cap \text{Spec } B$ is the union of all the open sets which are simultaneously distinguished open subschemes of $\text{Spec } A$ and $\text{Spec } B$.

Proof

We need to show that if $p \in \text{Spec } A \cap \text{Spec } B$, then there are $f \in A, g \in B$ such $p \in D_A(f) = D_B(g)$.

Let $f \in A$ be such that $\text{Spec } A_f$ is a distinguished neighborhood of p in $\text{Spec } A \cap \text{Spec } B$. And let $\text{Spec } B_g$ be a distinguished neighborhood of $\text{Spec } B$ in $\text{Spec } A_f$ containing p . Then $g \in \Gamma(\text{Spec } B, \mathcal{O}_X)$ restricts to an

element $g' \in \Gamma(\text{Spec } A_f, \mathcal{O}_X) = A_f$. Now

$$\text{Spec } B_g = \text{Spec } A_f - \{\mathfrak{p} \mid g' \in \mathfrak{p}\} = \text{Spec}(A_f)_{g'}.$$

We can write $g' = g''/f^n$ for $g'' \in A$ and then $\text{Spec}(A_f)_{g'} = \text{Spec } A_{fg''}$. Thus

$$\text{Spec } B_g = \text{Spec } A_{fg''}$$

is a neighborhood of p which is distinguished both in $\text{Spec } A$ and $\text{Spec } B$. ■

Lemma 2.5.23 (Affine communication lemma)

Let P be some property satisfied by some affine open subsets of a scheme X , such that for any affine open subset $\text{Spec } A \hookrightarrow X$,

- (1) if $\text{Spec } A \hookrightarrow X$ has property P , then for any $f \in A$, $\text{Spec } A_f \hookrightarrow X$ has P as well;
- (2) if $(f_1, \dots, f_n) = A$, and $\text{Spec } A_{f_i} \hookrightarrow X$ has P for all i , then so does $\text{Spec } A \hookrightarrow X$.

Now suppose that $X = \bigcup_i \text{Spec } A_i$ and $\text{Spec } A_i$ all have property P . Then *every* affine open subset of X has P .

Proof

Let $\text{Spec } A \hookrightarrow X$ be an affine open subscheme. Then we can cover $\text{Spec } A$ with a finite number of distinguished open subsets $\text{Spec } A_{g_i}$, which are each distinguished in some $\text{Spec } A_i$. This is possible by the previous proposition. As these are all distinguished in $\text{Spec } A_i$ which has P , by (1) each $\text{Spec } A_{g_i}$ has P . Then by (2) so too does $\text{Spec } A$. ■

Note:

An **affine local** property is a property which can be checked on any affine cover. Thus a property which satisfies the above two conditions is affine local. Note that if $U \subseteq X$ is an open subscheme, then U inherits any affine local property X has (because affine subschemes of U are affine in X).

Furthermore note that any stalk local property (a scheme has P iff every stalk has property Q) is affine local in the sense that the scheme has property P iff an affine cover has property R (the property R being that the affine scheme has stalks satisfying Q).

Definition 2.5.24

A scheme X is **locally Noetherian** if it can be covered by affine open sets $\text{Spec } A$ where A is Noetherian. And it is a **Noetherian scheme** if it is also quasicompact, or equivalently can be covered by finitely many affine open sets $\text{Spec } A$ where A is Noetherian.

By corollary 2.5.5 a Noetherian scheme has an underlying Noetherian topological space. The converse is in general not true; it is possible for $\text{Spec } A$ to be a Noetherian space without A itself being Noetherian.

A locally Noetherian scheme X is quasiseparated: take a cover by Noetherian affine schemes, then intersection of any two is then Noetherian (any subspace of a Noetherian space is Noetherian) and in particular quasicompact. Thus a Noetherian scheme is qcqs.

A Noetherian space has a finite number of irreducible components by proposition 2.1.16. Because every connected component is the union of irreducible components, a Noetherian space also has finitely many connected components each of which is a finite union of irreducible components.

Lemma 2.5.25

For a scheme X , the minimal prime ideals of the stalk $\mathcal{O}_{X,p}$ are in bijection with the irreducible components of X containing p .

Proof

We know this is true for affine schemes: irreducible subsets of $\text{Spec } A$ are the vanishing sets of prime ideals, and they are maximal precisely when the prime ideal is minimal.

Now if $U \subseteq X$ is open, then there is a bijection between irreducible components of U containing p and irreducible components of X containing p . Indeed if $p \in Z \subseteq U$ is an irreducible component, then $\overline{Z} = \text{cl}_X(Z)$ is irreducible. And if $p \in Y \subseteq X$ is an irreducible component then $Y \cap U \subseteq U$ is irreducible (open subsets of irreducible spaces are irreducible).

Note that $Y = \overline{Y \cap U} \cup (Y - U)$ are open; since Y is irreducible one must be Y . Since $p \notin Y - U$, we have that $\overline{Y \cap U} = Y$. And $\overline{Z} \cap U = \text{cl}_U(Z) = Z$ because Z is closed; so these are mutual inverses between closed irreducible subsets of U and of X containing p . These also preserve order, so they preserve maximal elements, i.e. irreducible components.

Thus for $p \in \text{Spec } A \subseteq X$, there is a bijection between irreducible components of X containing p and irreducible components of $\text{Spec } A$ containing p . The latter is in bijection with minimal prime ideals of $\mathcal{O}_{X,p}$. ■

Proposition 2.5.26

A nonempty Noetherian scheme X is integral iff X is connected and all of its stalks are integral.

Proof

In general an integral scheme X is irreducible and thus connected. Furthermore its stalks are all domains, as the direct colimit of domains is a domain: if $fg = 0$ then $f|_U g|_U = 0$ in some neighborhood U , but then $\mathcal{O}_X(U)$ is a domain so $f|_U$ or $g|_U$ is zero in $\mathcal{O}_X(U)$, so f_p or g_p is zero in $\mathcal{O}_{X,p}$.

Conversely suppose that X is connected and all of its stalks are integral. Then X is reduced, so by proposition 2.5.19 we just need to show that X is irreducible. Now suppose that X has two distinct irreducible components Z, Z' . If $p \in Z \cap Z'$ then $\mathcal{O}_{X,p}$ is not a domain, as by the previous lemma $\mathcal{O}_{X,p}$ has multiple minimal prime ideals. A domain has a unique minimal prime ideal, 0.

Thus X is the disjoint union of its irreducible components. There are finitely many of these, and as they are all closed, they must then be open. But a connected space has no non-trivial clopen subsets, so X has a unique irreducible component and is therefore irreducible. ■

Definition 2.5.27

Let A be a ring. An **A -scheme** is a scheme X whose sections are all A -algebras and its restriction maps are A -algebra morphisms. That is, it is a $(\text{Alg}_A, \text{LocAlg}_A)$ -space (where LocAlg_A obviously means local A -algebras).

If X is an A -scheme which can be covered by affine open subschemes $\text{Spec } B_i$ where each B_i is a finitely generated A -algebra, then X is **locally of finite type** over A , or a **locally finite type A -scheme**. If X is furthermore quasicompact, then X is of **finite type** over A , or a **finite type A -scheme**.

Recall that if A is Noetherian and B is a finitely generated A -algebra, then B is Noetherian as well (by Hilbert's basis theorem). Thus if X is a locally finite type A -scheme, then it is locally Noetherian. Similarly a finite type A -scheme is Noetherian.

For example $\mathbb{P}_A^n$ is a finite type A -scheme.

Definition 2.5.28

An affine scheme which is reduced and of finite type over a field k is said to be an **affine variety** over k . A reduced (quasi)projective k -scheme is a **(quasi)projective variety** over k .

Proposition 2.5.29

Suppose A is a ring and $(f_1, \dots, f_n) = A$.

- (i) if A is Noetherian, then so is A_{f_i} . Conversely if each A_{f_i} is Noetherian, so is A .
- (ii) suppose B is a ring and A is a B -algebra. Then A is a finitely generated B -algebra iff each A_{f_i} is.

Proof

For (1) we first note that the localization of a Noetherian ring is Noetherian. Recall that ideals of $S^{-1}A$ are all of the form $S^{-1}I$ for an ideal $I \leq A$. Now if we have a collection Σ of ideals of $S^{-1}A$, we can define $\Sigma' = \{I \leq A \mid S^{-1}I \in \Sigma\}$. Because A is Noetherian, this has a maximum element I , and then $S^{-1}I \in \Sigma$ is a maximum element.

Conversely if $I_1 \subseteq I_2 \subseteq \dots$ is an increasing chain of ideals in A , then let $I_{i,j} = (I_j)_{f_i}$. Thus $I_{i,1} \subseteq I_{i,2} \subseteq \dots$ is increasing in A_{f_i} and thus must stabilize. Now suppose that the $I_{i,j}$ all stabilize for $j \geq N$. Define $M = I_{j+1}/I_j$ for $j \geq N$, then M embeds in

$$M \hookrightarrow \prod_i M_{f_i} \cong \prod_i I_{i,j+1}/I_{i,j} \cong 0$$

so that $I_j = I_{j+1}$ as desired.

For (2) note that A_{f_i} is a finitely generated A algebra (by $1/f_i$), so if A is finitely generated then so is A_{f_i} . Conversely if $r_{ij}/f_i^{k_{ij}}$ generate A_{f_i} for $r_{ij} \in A$, and $1 = \sum_i c_i f_i$, then $\{f_i, c_i, r_{ij}\}$ generate A . ■

Combining (1) above with Affine communication lemma we get the following immediate corollary.

Corollary 2.5.30

A scheme X is locally Noetherian iff for every affine subscheme $\text{Spec } A \subseteq X$, A is Noetherian. In particular, an affine scheme $\text{Spec } A$ is a Noetherian scheme iff A is Noetherian.

Remark 2.5.31

Recall that $\text{Spec } A$ and $\text{Spec } A/\text{nil}(A)$ are homeomorphic as topological spaces.

Let $A = k[x_1, x_2, \dots]/(x_1^2, x_2^2, \dots)$. Then $\text{nil}(A) = (x_1, x_2, \dots)$ and $A/\text{nil}(A) \cong k$ so that $\text{Spec } A \cong \text{Spec } k$ is a one-point space and thus Noetherian. But $\text{nil}(A)$ is not a finitely generated ideal, so that A is not a Noetherian ring.

Thus $\text{Spec } A$ being a Noetherian space does not mean it is a Noetherian ring. In particular a scheme which has an underlying Noetherian space is not necessarily a Noetherian scheme.

2.5.4 Normality and factoriality

Recall that if B is an A -algebra, then an element $b \in B$ is **integral** over A if it is the root of some *monic* polynomial over A . It is a standard result that the set of all integral elements of B forms an A -subalgebra. We say that A is **integrally closed** in B if this subalgebra is A (more precisely, A 's image in B). That is, the only integral elements of B are those which are already in A .

An integral domain A is **normal** (or **integrally closed**) if it is integrally closed in its field of fractions $A \subseteq K(A)$.

Definition 2.5.32

A scheme X is **normal** if its stalks are all normal. That is, $\mathcal{O}_{X,p}$ is an integral domain and integrally closed in its fraction field.

Thus normal schemes are reduced (but not necessarily integral, as that is not a stalk-local property).

It is not hard to see that A is normal iff $A_{\mathfrak{p}}$ is for all prime $\mathfrak{p}$. Indeed $K(A_{\mathfrak{p}}) \cong K(A)$, and the intersection of the $A_{\mathfrak{p}}$ s in $K(A)$ is equal to A . In fact the intersection of the $A_{\mathfrak{m}}$ s in $K(A)$ is equal to A . Indeed, if $a/b \in K(A)$ is in $A_{\mathfrak{m}}$ then $b \notin \mathfrak{m}$, thus

$$\frac{a}{b} \in \bigcap_{\mathfrak{m}} A_{\mathfrak{m}} \iff b \notin \bigcup_{\mathfrak{m}} \mathfrak{m} \iff b \in A^{\times},$$

in which case $a/b \in A$. So if $f \in K(A)$ is integral over A , it is integral over each $A_{\mathfrak{p}}$, thus in each $A_{\mathfrak{p}}$ and so in A . Conversely the localization of a normal domain is normal.

Because the stalks of $\text{Spec } A$ are $A_{\mathfrak{p}}$, $\text{Spec } A$ is a normal scheme iff A is normal.

Remark 2.5.33

Recall that stalk-local properties are affine-local. Thus the following are equivalent:

- (i) X is normal,
- (ii) for every affine subscheme $\text{Spec } A \subseteq X$, A is normal,
- (iii) there exists an affine cover $X = \bigcup_i \text{Spec } A_i$ such that A_i is normal for all i .

Proposition 2.5.34

A scheme X is normal and Noetherian iff it is the finite disjoint union of integral Noetherian normal schemes.

Proof

Suppose X is a normal Noetherian scheme. Then let $X = \bigsqcup_{i=1}^n X_i$ be its connected components. These are closed subsets of X , thus quasicompact, and because X is locally Noetherian so too are the X_i . Thus each X_i is a connected Noetherian scheme. Furthermore the stalks of X_i are stalks of X , which are normal. Thus by proposition 2.5.26, each X_i is integral.

Thus $X = \bigsqcup_{i=1}^n X_i$ is a finite disjoint union of integral Noetherian normal schemes, as desired.

Conversely if $X = \bigsqcup_{i=1}^n X_i$ is a finite disjoint union of integral Noetherian normal schemes, then X is locally Noetherian by covering each X_i with a Noetherian affine cover. Because the X_i are all Noetherian, so too is X ; thus X is a Noetherian scheme. The stalks of X are stalks of X_i which are all normal, so X is a normal Noetherian scheme. ■

Recall that an integral domain A is a **unique factorization domain (UFD)** if every nonzero element $a \in A$ can be written uniquely as a product of irreducible elements $a = p_1 \cdots p_n$. Recall that an irreducible element $p \in A$ is an element that cannot be written as a nontrivial product: if $p = ab$, then one of a, b must be a unit.

Definition 2.5.35

A scheme X is **factorial** if all of its stalks are UFDs.

A UFD is also normal. The proof follows essentially the same as the classical one for $\mathbb{Z}$. If $p/q \in K(A)$ is integral over A , then suppose it is the root of $x^n + a_{n-1}x^{n-1} + \cdots + a_0$. We may assume, as A is a UFD, that

p, q are coprime. Plugging in $x = p/q$ and multiplying by q^n gives us

$$p^n + a_{n-1}p^{n-1}q + \cdots + a_0q^n = 0$$

which implies that q divides p , which means that q is a unit (as p, q are coprime). Thus $p/q \in A$ as desired. Thus factorial schemes are also normal.

Lemma 2.5.36

If A is a UFD then so too are all of its localizations $S^{-1}A$.

Proof

Firstly, $S^{-1}A$ is an integral domain.

If $p \in A$ is irreducible, then it is irreducible in $S^{-1}A$. Indeed, if $p = (ab)/s$ then $ps = ab$ and so p divides ab . Since p is irreducible and A is a UFD, it is prime, so suppose it then divides a . Then it divides a/s , so p is prime and thus irreducible in $S^{-1}A$.

Conversely if $p/s \in S^{-1}A$ is irreducible, then because $1/s$ is a unit, $p \in S^{-1}A$ is irreducible. But since A embeds in $S^{-1}A$, this means that $p \in A$ is irreducible. Thus, up to multiplication by a unit, the irreducible elements of $S^{-1}A$ are those of A .

Then any $a \in A$ has a unique factorization in $S^{-1}A$ by irreducibles, and thus $a/s \in S^{-1}A$ does as well. ■

In particular if A is a UFD, $\text{Spec } A$ is factorial.

Note:

The converse is not true, though. Even if $\text{Spec } A$ is factorial, A need not be a UFD.

Remark 2.5.37

UFDs can arise from some common constructions. Firstly, every PID is a UFD.

Indeed if A is a PID and $a \in A$ is nonzero, then if a is not irreducible we have $a = a_{1,1}a_{1,2}$ for non-units. If these are not irreducible, we can continue with $a = a_{2,1}a_{2,2}a_{2,3}a_{2,4}$. And so on, we get an ascending chain of ideals

$$(a) \subsetneq (a_{1,1}, a_{1,2}) \subsetneq \cdots \subsetneq (a_{i,1}, \dots, a_{i,2^i}) \subsetneq \cdots$$

But A is Noetherian, a contradiction.

And in a PID, irreducible elements are prime. Indeed, if $p \in A$ is irreducible then $(p) \leq A$ is maximal. Since if $(p) \subseteq (a)$ then $p = ab$ so either a is a unit or b is; so $(a) = A$ or $(p) = (a)$. Thus (p) is prime which means that if p divides ab , p divides either a or b , meaning p is prime.

Thus if $p_1 \cdots p_n = q_1 \cdots q_m$, then p_1 divides one of the q_i s, which means they are equal (up to unit), and so on. Thus A is a UFD.

Thus if k is a field, $k[x]$ is a PID and thus a UFD. In more generality it is true that if A is a UFD then $A[x]$ is a UFD.

Thus for example $\mathbb{A}_k^n, \mathbb{P}_k^n$ are factorial, as $\mathbb{A}_k^n = \text{Spec } k[x_1, \dots, x_n]$ where the polynomial algebra is a UFD. And $\mathbb{P}_k^n$ can be covered by $\mathbb{A}_k^{n+1}$ s. For the same reason, we can replace k with any UFD A .